Modern Geometric Reconstruction of the Luoshu Master Equation: Fiber Bundles, Resonance K, and Fractal Center Point 5

— Yin-Yang Fractal Unified Field Theory, Volume IV

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Abstract

This volume is Volume IV of the Yin-Yang Fractal Unified Field Theory, titled "Modern Geometric Reconstruction of the Luoshu Master Equation: Fiber Bundles, Resonance K, and Fractal Center Point 5."

Volume III established the Luoshu Master Equation, unified the four fundamental forces, and verified the masses of 33 particles. However, V3 left three unresolved theoretical gaps: (1) the ontology of Center Point 5 was ambiguous, with an antinomy between the global topological origin and the local fractal centers at various scales; (2) the resonance parameter K carried a conceptual ambiguity, serving both as geometric layer number and as resonance condition, lacking a unified topological definition; (3) the gravitational channel misapplied the empirical decay factor $(\Phi^5)^{38.32}$ of the longitudinal umbilical cord, lacking rigorous topological derivation.

V4's positioning is geometric reconstruction, not paradigm overthrow. All experimental comparison results of V3 that have passed numerical verification are retained. V4's task is to elevate V3's phenomenological and algebraic models to the modern mathematical physics framework of fiber bundles, differential geometry, and topology.

The core work of V4 consists of three parts. First, distillation: systematizing the core conclusions of V1–V3. Second, correction: correcting the three theoretical gaps of V3. Third, translation: translating the physical images of V1–V3 into the language of modern mathematical physics.

The core results of V4 are as follows. Center Point 5 is fractal: globally there is only the base-space zero point $n = 0$; the "local Center Point 5" seen at each scale is the local trivialization projection of the global zero point at that scale. K is simultaneously the generalized first Chern number and the resonance condition; two structures share K if and only if their frequency ratio is rational. The electron is below the atomic nucleus; containment does not equal revolution; the electron revolves around the atomic nucleus, not around Planck Center Point 5. The ionization energy is $E = P_0 \times \sum A_i \times (\Phi^5)^{d_i}$. The molecular bond energy is $E_{\text{bond}} = P_0^{(\text{mol})} \times a_A \times a_B \times \Phi^{-K_{\text{shared}}}$. The proton-proton gravitational force is $F = 1.05 \times 10^{-34}$ N, with a deviation of -1.9% from the experimental value. All formulas retain the form and numerical results of V3.

The theoretical framework is as follows. The universe is a fractal fiber bundle; the base space is the layer axis, amplified by 24 times per layer; the fiber is the phase circle S^1; the total space $E = B \times F$ constitutes a trivial fiber bundle. Spiral motion is equivalent to a fiber-bundle section; the particle is the topological jamming of the spiral wave on the Luoshu grid. Particle mass is the intersection number of submanifold transverse intersections, $M \propto I \cdot M_0$. The four fundamental forces are unified under topological shadow dynamics; the difference among the four forces is a difference in topological structure, not a difference in spacetime and fields. Gravitational weakness originates from the scarcity of topological degrees of freedom, and its channel is the universe's only perfectly lossless transverse closed fractal ring. Quantum mechanics and relativity are limit approximations of the fractal fiber bundle; the quantum gravity puzzle exists because two effective theories are mistaken for fundamental theories. The universe is a finite boundaryless, energy-circulating, dynamic steady state; a star is an aggregate of countless locally activated secondary vortex sources; dark matter is the gravitational projection of adjacent-dimensional cosmic webs; dark energy is the energy-particle background pressure ejected by countless stellar generators.

V4 also presents a series of observable exclusive predictions, including a fourth-generation lepton mass of approximately 8.60 GeV, ionization energies of period-8 p-block elements at 6.19–10.98 eV, an ionization energy of 6.0 ± 1.0 eV at the island-of-stability center $Z = 126$, a tiny excess radiation from stellar generators beyond nuclear fusion, and a dependence of galactic gravitational anomalies on topological morphology.

Keywords: Yin-Yang fractal; Luoshu grid; golden ratio; breathing; spiral motion; particle mass spectrum; four fundamental forces; shadow dynamics; fractal hierarchy; fiber bundle; generalized Chern number; resonance K

Introduction

0.1 The Theoretical Construction of V1–V3

V1–V3 completed the theoretical construction centered on the Luoshu Master Equation, obtaining 33 particle mass spectra, phenomenological descriptions of the four fundamental forces, atomic ionization energies, molecular bond energy calculation results, and preliminary connections between quantum mechanics and relativity.

Running through this series of work is a physical image that penetrates all scales. These images are the most original source of thought: the reason the mass formulas can achieve numerical precision is that the underlying images are correct. This volume continues the general principle of "image first, formula follows."

0.2 Three Key Theoretical Gaps Left by V3

1. **Ambiguity in the ontology of Center Point 5**: an antinomy exists between the global topological origin and the local fractal centers at various scales.
2. **Conceptual ambiguity of the resonance parameter K**: K simultaneously serves as geometric layer number and resonance condition, lacking a unified topological definition.
3. **Error in the decay factor of the gravitational channel**: the empirical decay factor $\left(\Phi^5\right)^{38.32}$ of the longitudinal umbilical cord was misapplied, lacking rigorous topological derivation.

0.3 V4 Positioning

V4 positioning: **geometric reconstruction, not paradigm overthrow**. All experimental comparison results of V3 that have passed numerical verification are retained; the work of V4 is to elevate the phenomenological and algebraic models of V3 to the modern mathematical physics framework of fiber bundles- differential geometry- topology.

The core work of V4 includes three parts:

1. **Distillation**: systematizing the core conclusions of V1-V3.
2. **Correction**: correcting the three theoretical gaps of V3.
3. **Translation**: translating the physical images of V1-V3 into the language of modern mathematical physics.

V4 is an abbreviated version. The original derivations, image details, and complete numerical verification processes are in V1 (35 volumes). Readers are advised to read them in comparison.

0.4 Core Conclusions of V4

1. Center Point 5 is fractal. Globally there is only the base space zero point $n = 0$. The "local Center Point 5" seen at each scale is the local trivialization projection of the global zero point at that scale.
2. K is simultaneously the generalized Chern number and the resonance condition.
3. The electron is below the atomic nucleus. Containment does not equal revolution. The electron revolves around the atomic nucleus, not around Planck Center Point 5.
4. Ionization energy $= P_0 \times \sum A_i \times \left(\Phi^5\right)^{d_i}$.
5. Molecular bond energy $= P_0^{(\mathrm{mol})} \times a_A \times a_B \times \Phi^{-K_{\mathrm{shared}}}$.

6. Proton-proton gravity: $F = 1.05 \times 10^{-34}$ N, deviation from experimental value -1.9%.
7. All formulas retain the form and numerical results of V3.

Symbol Conventions

- $\Phi = \frac{\sqrt{5}-1}{2} \approx 0.618034$, the golden ratio; $\Phi^5 \approx 0.09017$
- ℓ_P: Planck length; t_P: Planck time
- B: base space (layer axis); $F = S^1$: phase circle fiber; $E = B \times F$: trivial fiber bundle total space
- K: generalized first Chern number (resonance K). K is simultaneously the generalized Chern number and the resonance condition: two structures share K if and only if their frequency ratio is rational. The resonance essence of K: $\omega_K = \omega_0 \times 24^{-K}$.
- $W[K] = 1/\Phi^K$: topological channel weight
- $\mathrm{Hol}(\gamma)$: closed-path holonomy
- $I(M_1, M_2)$: submanifold transverse intersection number
- $P_0^{(\mathrm{local})}$: local cosmic web pressure
- η: shadow interception rate; A_{eff}: effective geometric cross-section

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Chapter 1 Rigorous Fiber-Bundle Definition of Center Point 5

1.0 Chapter Positioning

This chapter is the axiomatic layer and mathematical foundation layer of the entire book, resolving the ontological antinomy of Center Point 5 left by V3. The writing follows the general principle of the whole book: image first, formula follows. First inherit the physical intuition of V3's breathing- convex- concave- rotation- fountain, then transition to the rigorous mathematical definition of fiber bundles.

1.1 Physical Image: Center Point 5 — the Source of Breathing

Before entering abstract mathematical tools, completely inherit the core dynamical image of V3's [Breathing Axiom (Axiom D)].

Center Point 5 is a topological source that continuously and steadily ejects primordial energy particles outward. It is not the celestial singularity of violent collapse or instantaneous big bang in general relativity. Its most fundamental form of motion is breathing motion: alternating convex-concave oscillation.

- **Expansion phase (convex)**: ejects primordial energy particles outward;
- **Contraction phase (concave)**: curls inward, giving rise to vortex structures.

A single convex-concave reciprocating motion only produces a purely radial in-and-out, without producing rotation. Only when multiple convex-concave motions have phase differences with each other, and are temporally staggered, does rotation arise; the trajectory of rotation is the spiral.

From this global vortex source at Center Point 5, three mutually orthogonal spiral rings are generated: the corner ring $2 \to 4 \to 8 \to 6 \to 2$, the vertical edge-midpoint ring $1 \to 3 \to 9 \to 7 \to 1$, and the horizontal edge-midpoint ring $1 \to 3 \to 9 \to 7 \to 1$. The three spiral rings mutually penetrate and support each other; their projection superposition constitutes the three-dimensional skeleton of the Luoshu nine-grid.

From the perspective of fluid topology: Center Point 5 is equivalent to the global vortex source of vortex theory, continuously generating vortex tubes; this theory calls them umbilical cords. The Helmholtz- Moffatt theorem guarantees the topological stability of vortex tubes: a vortex tube cannot be interrupted in the interior of a fluid out of nowhere, so the umbilical cord cannot be cut by continuous small perturbations.

1.2 Fractal Nesting and Stars as Local Secondary Vortex Sources

Fractal structure possesses nested containment relations. The interior space of massive celestial bodies (stars, planets, atomic nuclei) all contains countless submanifold units at the Planck scale. These sub-units are large numbers of local trivialization projections of the global Center Point 5 at higher scales.

Key distinction: "containment" does not equal "replicating the global origin," nor does it mean naturally becoming a generator.

Different-scale celestial bodies have different degrees of topological activation of the internal Planck-scale sub-units, divided into two categories:

First category: topologically unactivated objects (rocks, planets). Although their interiors contain Planck-scale sub-units, they do not reach the activation threshold of vortex topology. They can only receive energy particles, not on a large scale newly generate and eject them.

Second category: topologically activated objects (stars). The core region of a star possesses extremely high temperature and strong vortex topological conditions, topologically activating large numbers of dormant Planck-scale sub-units in its interior. From the microscopic level, countless local secondary vortex sources are ejecting energy particles; rising to the macroscopic observational perspective, this manifests as the star's overall oscillatory ejection — the star's star-making motion.

A star is not a single generator; it is an aggregate of countless activated local secondary vortex sources. A star is a macroscopic-scale composite secondary vortex source activated by topology. A star continuously newly generates and ejects energy-substance particles, participating in weaving the cosmic web.

Hierarchical relations:

- Global Center Point 5: the universe's unique ultimate vortex source, located at base space $n = 0$;
- Star: an aggregate of countless local secondary vortex sources activated by topology, located at the stellar hierarchy;
- Planets, rocks: unactivated passive receiving objects.

The three constitute a fractal nesting relation of "global source- nested activated local source aggregate- passive receiving object."

1.3 Base Space B: Layer Axis

Definition 1.3.1 (Base Space)

The base space B is the layer axis, a one-dimensional manifold, whose parameter marks the scale hierarchy of the fractal structure, not the radial coordinate of ordinary Euclidean space:

$$B = \{n \in \mathbb{R} \mid 0 \leq n \leq n_{\max}\}$$

Key characteristic layers:

- $n = 0$: Planck scale, global Center Point 5, the base space zero point of the fiber bundle;
- $n = 16.79$: the standing-wave layer where the electron undergoes topological jamming (lepton stable standing-wave layer, not a center point coordinate);
- $n = 19.16$: proton layer, hadron stable standing-wave layer;
- $n = 56.5$: Earth layer, planetary fractal resonance layer;
- $n = 60.5$: Sun layer, stellar fractal resonance layer.

Base space metric. Because of fractal self-similarity, each rise of one layer amplifies the spatial characteristic scale by 24 times; the base space line element is taken as:

$$ds_B^2 = (\ln 24)^2 dn^2$$

This metric guarantees: the geometric distance from $n = 0$ to $n = 1$ is equivalent to that from $n = 1$ to $n = 2$, embodying the translational symmetry (scale invariance) of the fractal.

1.4 Fiber F: Phase Circle

Definition 1.4.1 (Fiber)

Above each layer point n of the base space, a fiber describing the spiral rotation phase is attached, and the fiber is the unit phase circle:

$$F = S^1 = \{\theta \mid 0 \leq \theta < 2\pi\}$$

The fiber describes the rotation phase of spiral motion, not representing ordinary spatial position.

The spiral characteristic radius of the n-th layer:

$$r_n = \frac{\ell_P \cdot 24^n}{2\pi}$$

The fiber metric describes the physical arc length corresponding to a phase change on the same layer:

$$ds_F^2 = r_n^2 d\theta^2$$

Physical meaning: at higher layers where $n$is larger, the same phase change $d\theta$corresponds to a larger actual spatial arc length; this is the geometric expression of the spiral unfolding outward.

1.5 Total Space E: Trivial Fiber Bundle

Definition 1.5.1 (Total Space Fiber Bundle)

Taking the Cartesian product of base space B and phase fiber F gives the total space describing the entire fractal universe:

$$E = B \times F$$

The projection map $\pi : E \to B$is defined as $\pi(n, \theta) = n$.

πprojects the point (n, θ)in the total space to the layer number $n$in the base space. (E, B, π, F)constitutes a trivial fiber bundle. For any layer number n, the fiber on that layer is the preimage $\pi^{-1}(n) = \{n\} \times F$.

Local trivialization. In any local neighborhood $U \subset B$, the fiber bundle is locally homeomorphic to $U \times F$. The "local Center Point 5" observed at each scale corresponds to a set of local trivialization coordinates. Different scales are connected by transition functions; this is the mathematical carrier of fractal self-similarity.

1.6 Spiral: Section of the Fiber Bundle

Definition 1.6.1 (Section)

A section is a map $s : B \to E$, satisfying the composition condition:

$$\pi \circ s = \mathrm{id}_B$$

Physical meaning: for each layer number $n$of the base space, the section $s$assigns a unique phase $\theta(n)$. Spiral motion is a section of the fiber bundle.

The spiral section parameter equation corresponds to the V3 geometric law of "five steps one circle":

$$\theta(n) = \theta_0 + 2\pi\frac{n}{5}$$

Physical meaning: when the layer axis advances $\Delta n = 5$, the phase increases exactly by 2π; the spiral completes a full circle. All spiral sections start from the base space zero point $n = 0$, i.e., in the physical image, all spirals eject outward from the vortex source at Center Point 5.

1.7 Theorem: Global Center Point 5 Is the Unique Zero Point n=0 of the Base Space

Theorem: The base space B of the fiber bundle has only one global zero point $n = 0$; all spiral sections $s$take this point as their common starting point. Therefore, global Center Point 5 is the unique topological origin (single-particle center at the Planck scale).

Proof sketch: The base space B is a one-dimensional layer axis, and only has the unique coordinate zero point $n = 0$; the initial condition constraints of all spiral sections require $n = 0$to be the common starting point of the phases. Therefore, global Center Point 5 exists and is unique.

Clarification of the old ambiguity: The so-called "each layer has its own Center Point 5" does not mean that the global origin has moved to a higher layer, nor that the particle center is located at electron layer 16.79.

$n_e = 16.79$is the layer number of the scale at which, after the spiral propagates outward from $n = 0$, the spiral wave undergoes topological jamming and forms the electron standing wave; it is strictly derived from the fractal grid edge length equaling the electron Compton wavelength:

$$n_e = \log_{24}\left(\frac{\lambda_e}{\ell_P}\right) \approx 16.79$$

The standing-wave phenomenon occurs at $n = 16.79$, but the topological source and true ontological center of the particle is still Planck Center Point 5 at base space $n = 0$. The local center seen at each scale is the projection of the global zero point $n = 0$under the corresponding local trivialization.

Definition 1.7.1 (Fractal Center Point 5)

Denote by $C_m$the fractal local Center Point 5 corresponding to the m-th scale (local trivialization projection):

- C_0 : Planck Center Point 5, corresponding to $n = 0$(global topological origin, the true source of all spirals, single-particle center);
- C_1 : the local projection Center Point 5 at the particle scale (the local coordinate system origin at the electron scale, projected from C_0 , not equal to $n = 16.79$; $n = 16.79$is the layer at which standing-wave locking occurs);
- C_2 : atomic Center Point 5, corresponding to the local projection at the atomic scale;
- C_3 : stellar Center Point 5, corresponding to the local projection at the stellar scale $n = 60.5$.

Nested containment relation:

$$C_0 \subset C_1 \subset C_2 \subset C_3 \cdots$$

Important conceptual distinction: containment ≠ revolution.

Containment: set/topological subset embedding relation; represents that a higher-level celestial body internally accommodates lower-level fractal units.

Revolution: dynamical orbital periodic motion.

The two belong to completely orthogonal concepts.

- The fact that a star internally contains Planck-scale sub-units does not mean that the star dynamically orbits around the Planck-scale origin. The star's own revolution is a dynamical behavior relative to the center of a higher-level celestial body.
- The electron revolves around the atomic nucleus, but the electron does not revolve around Planck Center Point 5. Planck Center Point 5 is the base space origin, not a dynamical revolution object.

1.8 Electron: Topological Jamming of the Spiral Wave on the Luoshu Grid

Definition 1.8.1 (Topological Jamming)

The spiral wave propagates outward along the fiber-bundle section from the global origin $n = 0$; when the spiral phase and the Luoshu discrete grid phase satisfy:

$$\theta_{\text{spiral}}(n) = \theta_{\text{grid}}(n) + C$$

where C is a constant, then the spiral wave is said to undergo topological jamming.

Physical interpretation: this is a topological phase constraint, not the binding of a mechanical potential well. The spiral wave propagates outward to the layer $n_e = 16.79$, satisfies the standing-wave condition, and is locked by the discrete grid phase condition, forming a stable standing wave, i.e., the electron particle.

Jamming occurs at $n_e = 16.79$, but the topological source center of the particle is still Planck Center Point 5 at $n = 0$. Only when a topological phase transition occurs (particle annihilation or energy-level transition) can the lock be released.

Electron standing-wave matching condition (reproduction of the original V3 derivation):

Fractal grid edge length formula:

$$\ell_n = \ell_P \cdot 24^n$$

Standing-wave locking condition: the Luoshu grid edge length of that layer equals the electron Compton wavelength $\lambda_e = h/(m_e c) \approx 2.42631 \times 10^{-12}\text{m}$:

$$\ell_{16.79} = \lambda_e$$

Solving for the layer number:

$$n_e = \log_{24}\left(\frac{\lambda_e}{\ell_P}\right) = \frac{\ln(\lambda_e/\ell_P)}{\ln 24} \approx \frac{\ln(1.5012\times10^{23})}{\ln 24} \approx \frac{53.39}{3.178} \approx 16.79$$

Standing-wave geometric condition:

$$2\pi r_e = m \cdot \ell_{16.79}$$

$r_e$is the electron spiral radius; $\ell_{16.79}$is the Luoshu grid edge length at layer $n = 16.79$; $m$is an integer. In the ground state $m = 1$, the electron forms the most stable standing wave.

The topological essence of jamming. Jamming is not mechanical; it is topological. It is not "jammed by force" but "phase-locked." Phase locking is a topological invariant. Once locked, it cannot be unlocked by continuous deformation.

1.9 The Relationship Between the Electron, Planck Center Point 5, and the Atomic Nucleus

This section focuses on clarifying the complete relationship between the electron and Planck Center Point 5 and the atomic nucleus. This is one of the core ambiguities left by V3.

(1) Motions in which the electron participates.

- **Source relation**: The electron is the topological jamming of the spiral wave at layer $n = 16.79$. The spiral wave starts from global Center Point 5 ($n = 0$). Therefore, the source of the electron is Planck Center Point 5.
- **Dynamical revolution**: Inside the atom, the electron revolves around the atomic nucleus. This is the electron's dynamical revolution object.
- **Phase reference**: The breathing vortex of Planck Center Point 5 provides a global phase reference, modulating the electron's orbit.

(2) The electron does not revolve around Planck Center Point 5.

The electron revolves around the atomic nucleus, not around Planck Center Point 5. Planck Center Point 5 is the electron's topological source, not its dynamical revolution object. "Comes from" and "revolves around" are two different things.

(3) Double-vortex intersection.

The electron is stably located at the intersection of the "Planck vortex phase" and the "atomic-nucleus vortex phase." The intersection condition is phase matching:

$$\Delta\phi = \phi_P - \phi_A = \pm\frac{\pi}{2}$$

But note: intersection is phase matching, not dynamical revolution. The Planck vortex is a background field providing phase reference, not revolution.

(4) The electron is below the atomic nucleus.

The electron layer $n_e = 16.79$is smaller than the atomic-nucleus layer $n_p = 19.16$. The electron is located below the atomic nucleus on the layer axis. The electron's depth relative to the atomic nucleus:

$$d_{\text{free}} = n_p - n_e = 19.16 - 16.79 = 2.37$$

This relative depth is the starting point for the ionization energy calculation.

(5) Comparison with the Earth-Sun analogy.

The Earth revolves around the Sun; the Solar System revolves around the galactic center. Does the Earth participate in two revolutions? No. The Earth only revolves around the Sun. The gravitational field of the galactic center influences the motion of the Solar System, but the Earth does not directly orbit the galactic center. The galactic center is a background field, not the Earth's revolution object.

The relationship between the electron and Planck Center Point 5 is the same: Planck Center Point 5 is a background field, not the electron's revolution object.

1.10 Umbilical Cord: Connection of the Fiber Bundle

Definition 1.10.1 (Connection)

A connection is a 1-form on the fiber bundle that specifies the rule for parallel transport between different fibers. In this theory, the umbilical cord is the connection.

$$A = A_n dn$$

The connection coefficient:

$$A_n = \ln \frac{1}{\Phi^5} \approx 2.406$$

The umbilical cord is the line integral of the connection along the path γ:

$$\int_\gamma A = \int_0^{n_p} A_n dn = A_n \cdot n_p$$

Physical meaning: represents the accumulation of phase and energy on the longitudinal umbilical-cord channel from global Center Point 5 to a higher standing-wave layer.

Key distinction: The longitudinal umbilical-cord channel retains the Φ^5 decay effect; the transverse closed topological channel corresponding to gravity does not use this decay. This is a key distinction of V4, to be derived in detail in Chapter 7.

1.11 Chapter Summary

- The physical image inherits V3's breathing axiom: Center Point 5 is the source of breathing, the phase difference of convex-concave produces rotation, and rotation generates the spiral. Center Point 5 is the universe's unique ultimate vortex source, located at base space $n = 0$.
- **A star is not a single generator; it is an aggregate of countless activated local secondary vortex sources.** A star's interior contains large numbers of Planck-scale sub-units; after strong vortex topological activation, they become macroscopic-scale

composite secondary vortex sources, manifesting as star-making motion. Planets and rocks are not activated and can only passively receive.

- The base space B is the layer axis, with metric $ds_B^2 = (\ln 24)^2 dn^2$, describing the fractal scale hierarchy.
- The fiber $F = S^1$ is the phase circle; the total space $E = B \times F$ constitutes a trivial fiber bundle.
- Spiral motion is equivalent to a fiber-bundle section; five steps one circle is described by the section parameter equation $\theta(n) = \theta_0 + 2\pi n/5$.
- Global Center Point 5 is the unique zero point $n = 0$ of the base space; the fractal Center Point 5 at each scale is a local trivialization projection, satisfying the nested relation $C_0 \subset C_1 \subset C_2 \cdots$; strictly distinguish topological "containment" from dynamical "revolution."
- **The relationship between the electron and Planck Center Point 5 and the atomic nucleus:**
    - Source: the electron comes from Planck Center Point 5 (the spiral wave starts from $n = 0$);
    - Revolution: the electron revolves around the atomic nucleus;
    - Phase reference: the Planck vortex provides phase reference, modulating the electron orbit;
    - The electron does not revolve around Planck Center Point 5;
    - Double-vortex intersection is phase matching, not revolution;
    - The electron is below the atomic nucleus, with relative depth $d_{\text{free}} = 2.37$.
- The umbilical cord is the fiber-bundle connection; the Helmholtz- Moffatt theorem guarantees the topological stability of the umbilical cord.

One-sentence summary: The truly global Center Point 5 is forever at Planck $n = 0$; layer 16.79 is the layer where the spiral ran out and got jammed, generating the electron. A star is not a single generator but an aggregate of countless activated local secondary vortex sources. Containment does not equal revolution: the electron comes from Planck Center Point 5, but revolves around the atomic nucleus.

Chapter 2 Resonance K: Generalized Chern Number and Resonance Condition

2.0 Chapter Positioning

This chapter is the derivation layer and topological core layer of the entire book.

In the V3 version, the resonance parameter K exhibited conceptual ambiguity: it was used both as geometric layer number (describing the scale of fractal outward expansion) and as resonance condition (describing the phase-matching condition of spiral-grid coupling). These two sets of physical content shared the same symbol, lacking a unified topological mathematical definition, causing conceptual vacillation in the derivation process.

The core tasks of this chapter in V4:

4. Begin with the physical image of V3's "five steps one circle, regular pentagon golden phase unit";
5. Review the definition, mathematical properties, and physical limitations of the standard fiber-bundle first Chern number;
6. Introduce the generalized first Chern number (resonance K), allowing it to take real values, adapting to this theory's non-compact base space B (layer axis);
7. Establish K's dual identity: K is both the generalized Chern number (topological quantity) and the resonance condition (phase-matching criterion);
8. Define shared K: the criterion for local isomorphism between two physical systems' fiber bundles, providing the topological foundation for subsequent molecular bond energies and particle family resonance;
9. Discuss the splitting evolution law of K, compare with V3's empirical splitting $\Delta K_{m+1} = 2.164 \Delta K_m$, and distinguish the strictly proven parts from open problems requiring further analytical derivation;
10. Establish the direct connection between K (generalized Chern number) and channel weight $W[K] = 1/\Phi^K$, laying the groundwork for Chapter 5's four-force channel distribution ratios.

Preliminary conventions: All symbols from Chapter 1 are inherited. The base space $B = \{n \in \mathbb{R} \mid 0 \le n \le n_{\max}\}$ is a non-compact one-dimensional layer axis; the fiber $F = S^1$ is the phase circle; the total space $E = B \times F$ is a trivial fiber bundle; the spiral is a section $s : B \to E$, satisfying $\theta(n) = \theta_0 + 2\pi n/5$.

2.1 Physical Image First: Five Steps One Circle, Revolution-Rotation Phase Unit

Before entering the differential-geometric formulas, return to the physical image inherent in V3.

The spiral propagates outward along the base space from global Center Point 5 ($n = 0$), following the law of five steps one circle: the layer axis advances $\Delta n = 5$, the phase θ increases completely by 2π, and the spiral completes one revolution, its trajectory forming a regular pentagon under projection.

The ratio of the diagonal to the side of a regular pentagon is the golden ratio Φ.

Inside each macroscopic revolution step (72°), there are also 5 subdivided rotation phase units; each subdivided phase rotates $72^\circ/5 = 14.4^\circ$.

Revolution 5 steps 脴 rotation 5 phases = 25 phase positions; since the starting and ending phases are topologically coincident, the number of independent phases is $25 - 1 = 24$. This is the image source of the 24-fold amplification constant in Chapter 4.

This set of "revolution-rotation nested phase units" is precisely the physical object that resonance K seeks to describe: K is used to quantitatively describe the overall degree of distortion of the spiral fiber bundle, comprehensively including the number of revolution turns, the internal rotation subdivided phases, and the fractal scale coupling effect.

The standard differential-geometric first Chern number $c_1(E)$ is an integer, applicable to complex line bundles on compact closed manifolds. But this theory's base space B is a non-compact one-dimensional layer axis, which has no boundary closure; the layer number n can extend from 0 all the way to $n_{\max}$, the spiral section can extend infinitely outward, and the total degree of distortion of the spiral is not necessarily an integer number of turns. Therefore: directly copying the standard integer first Chern number cannot fully describe the Luoshu fractal spiral; generalization is needed to obtain a generalized first Chern number that can take real values, i.e., the resonance parameter K.

2.2 Standard First Chern Number: Definition, Properties, and Limitations for This Theory

2.2.1 Mathematical Definition (Standard Compact Complex Line Bundle)

Let $E \to M$ be a complex line bundle on a compact smooth manifold M. ω is the connection 1-form on the bundle; $\Omega = d\omega$ is the curvature 2-form.

Standard first Chern number:

$$c_1(E) = \frac{i}{2\pi} \int_M \Omega$$

Core properties:

4. $c_1(E)$ is necessarily an integer; it represents the total number of turns of the overall distortion of the fiber bundle on the closed base manifold;
5. Topological invariant: under continuous smooth deformation of the manifold, the value of $c_1(E)$ remains unchanged, independent of the metric and of the specific choice of connection, determined only by the topological type of the bundle;
6. Direct-sum bundle satisfies the additive property: $c_1(E_1 \oplus E_2) = c_1(E_1) + c_1(E_2)$.

2.2.2 Why the Standard $c_1(E)$ Cannot Be Directly Taken as This Theory's K

- The standard Chern number integration base M requires a compact closed manifold; this theory's base space B is a non-compact layer axis, with no closed boundary and no compact boundary for the integration domain;
- Physically, the Luoshu spiral extends infinitely outward, and the total distortion can be a real number, not necessarily an integer; for example, the electron $K_e = 16.79$, the muon $K_\mu = 14.55$, and the proton $K_p = 19.16$ are all non-integers;
- Therefore: K is not the standard first Chern number, but a generalized first Chern number.

The core idea of topological invariance is still inherited: even when generalized to real numbers, K remains a topological quantity characterizing the degree of distortion of the fiber bundle, unaffected by local continuous deformation.

2.3 Definition: Generalized First Chern Number (Resonance K)

2.3.1 Definition

Definition 2.3.1 (Generalized First Chern Number, Resonance K)

For the trivial fiber bundle $E = B \times S^1$ defined on the non-compact base space B (layer axis), take the connection ω and the curvature $\Omega = d\omega$. After integrating over a finite truncation interval $[0, n^*] \subset B$, take the limit extending to the entire base space:

$$K \equiv \lim_{n^* \to n_{\max}} \frac{i}{2\pi} \int_{[0,n^*]} \Omega$$

- K is called the resonance K (generalized first Chern number);
- K can take real values, no longer forced to be integer;
- Inherits topological invariance: under continuous smooth deformation of the fiber bundle, K remains unchanged; only when a topological phase transition occurs (vortex tube breaking, spiral annihilation, particle transition) does K change.

2.3.2 Physical Interpretation

- Larger K: represents a higher degree of overall distortion of the spiral fiber bundle;
- K directly encodes the "five steps one circle" revolution turns + internal rotation subdivided phase contributions;
- For particles: each type of stable particle corresponds to a definite set of K values:
    - Electron: $K_e = 16.79$
    - Muon: $K_\mu = 14.55$
    - Tau: $K_\tau = 13.64$
    - Proton: $K_p = 19.16$

Physical image restatement: K is not simply counting "how many layers n have been traveled" (geometric layer number); n is the base space coordinate parameter; K is the topological measure of distortion. For the same layer number n, if the internal rotation phase structure of the spiral changes, K can be different.

This resolves V3's ambiguity: n (geometric layer number, base space coordinate) and K (generalized Chern number, topological distortion) are strictly distinguished as two physical quantities.

When the base manifold is taken as a compact closed manifold, this definition automatically degenerates to the standard integer first Chern number $c_1(E)$. That is, the standard Chern number is a special case of the generalized K under the compact condition.

2.3.3 Direct-Sum Superposition Property

The generalized Chern number inherits the direct-sum additive rule:

$$K(E_1 \oplus E_2) = K(E_1) + K(E_2)$$

Physical interpretation: multiple spiral sub-bundles superpose, and the total resonance K equals the sum of the K's of the individual sub-bundles. This property is used later for quark composite baryons and mesons; the total topological distortion of a composite particle is obtained by superposing the K's of the submanifolds.

2.4 K's Dual Identity: Generalized Chern Number + Resonance Condition

This section is one of the core corrections of V4 relative to V3.

2.4.1 K Is Simultaneously the Generalized Chern Number and the Resonance Condition

V3 used K as a geometric layer number; this volume clarifies that K has a dual identity:

- **Identity one (topological quantity)**: K is the generalized first Chern number, characterizing the overall degree of distortion of the fiber bundle.
- **Identity two (resonance condition)**: K is the phase-matching criterion, characterizing whether two structures can undergo stable resonance.

The two identities are not two parallel things, but two descriptions of the same quantity. K's "degree of topological distortion" and "phase-matching condition" are in this theory two expressions of the same thing.

2.4.2 Mathematical Formulation of the Resonance Condition

Definition 2.4.1 (Resonance Condition)

Two structures A and B resonate if and only if their frequency ratio is rational:

$$r_{AB} = \frac{\omega_A}{\omega_B} = \frac{m}{n}$$

where m and n are integers. The frequency ratio being rational means the periods of the two structures have a common multiple and can be phase-locked.

Proposition (Equivalent Condition for Resonance)

$$r_{AB} = \frac{m}{n} \iff \Delta K = \log_{24}\left(\frac{m}{n}\right)$$

Resonance requires the layer-number difference to be a rational logarithm base 24.

2.4.3 Relationship Between Eigenfrequency and K

Let the characteristic time of the K-th layer be:

$$\tau_K = \tau_0 \cdot 24^K$$

The corresponding eigenfrequency is:

$$\omega_K = \frac{1}{\tau_K} = \omega_0 \cdot 24^{-K}$$

where $\omega_0 = 1/\tau_0$ is the Planck frequency. Larger K means lower frequency.

2.4.4 Shared K

Definition 2.4.2 (Shared K)

The fiber bundles corresponding to two physical systems are E_A and E_B respectively. If, in the local neighborhood of their mutual interaction, the fiber bundles are locally isomorphic $E_A|_U \cong E_B|_U$, then the two systems are said to share K. The necessary condition for local isomorphism of fiber bundles (and, under this theory, also the sufficient condition):

$$K_A = K_B$$

Physical interpretation:

1. Sharing K means the topological distortion degrees of the two spiral fiber bundles in the interaction region are exactly equal;
2. Only when $K_A = K_B$ is satisfied can the revolution-rotation phase units of the two spirals undergo perfect resonance matching;
3. Shared K possesses topological stability: as long as no topological phase transition occurs, continuous perturbations will not destroy $K_A = K_B$, explaining the dynamical stability of chemical bond structures;
4. In Chapter 8's molecular bond energy formula $E_{\text{bond}} = P_0^{(\text{mol})} \cdot a_A a_B \cdot \Phi^{-K_{\text{shared}}}$, K_{shared} is precisely the shared generalized Chern number here.

Shared K only requires local neighborhood isomorphism, not global isomorphism of the entire total-space fiber bundle. Atoms resonate only in the overlapping region of the boundary sphere; far from the bonding region the two bundles can be completely different.

2.4.5 Hierarchical Nesting of K

K is hierarchically nested:

$$K_1 \subset K_2 \subset K_3 \subset \cdots \subset K_n$$

Hierarchy	K value	Corresponding structure
K_0	0	Planck Center Point 5
K_1	K_1	Number-theory layer (twin primes)
K_2	K_2	Electron orbit layer
K_3	K_3	Nucleon layer
K_4	K_4	Atomic layer
K_5	K_5	Molecular layer
K_6	K_6	Protein/DNA layer
K_n	K_n	Cosmic layer

Proposition (Nesting Relation): If K_m and K_{m+1} are the K values of adjacent hierarchies, then $K_m \subset K_{m+1}$. Sharing K_n means sharing all lower-layer K's.

2.5 Splitting Evolution Law of K; Discussion of V3's Empirical Coefficient 2.164

2.5.1 Ideal Simple Case: Pure Revolution Turn Increase

If there is only revolution, then each complete revolution adds one complete set of phase units. In the ideal case:

$$K_{m+1} = K_m + 1$$

This corresponds to simply increasing the revolution turns, with the internal rotation phase structure unchanged.

2.5.2 V3 Empirical Splitting Law

But V3's numerical practice obtained the empirical splitting law:

$$\Delta K_{m+1} = 2.164 \cdot \Delta K_m$$

That is, the increment of K is amplified by a factor of 2.164, not simply +1.

Proposition: This empirical splitting $\Delta K_m = 2.164 \cdot \Delta K_{m-1}$originates from the contribution of higher-order fiber-bundle invariants (the second generalized Chern number c_2), rather than the simple turn increase of the first generalized Chern number.

2.5.3 First Chern Number vs Second Chern Number

- First generalized Chern number $K = c_1$: characterizes the one-dimensional distortion of the fiber along the base (phase rotation);
- Second Chern number $c_2(E) = \frac{1}{8\pi^2}\int_B \mathrm{tr}(\Omega \wedge \Omega)$: characterizes higher-order bidirectional distortion of the fiber bundle and internal coupling of the fiber.

The empirical amplification coefficient 2.164 observed in this paper comes from the correction contribution of the second generalized Chern number to the total K.

V3 obtained the empirical splitting relation $\Delta K_{m+1} = 2.164 \Delta K_m$through fitting a large number of particle spectra; this effect points to the higher-order topological contribution of the second generalized Chern number. This volume retains this empirical law for numerical verification; its complete analytical derivation is left as an open problem.

2.6 Direct Connection Between K and Channel Weight $W[K] = 1/\Phi^K$

In Chapter 1 we obtained the connection coefficient $A_n = \ln(1/\Phi^5)$; now we establish the relation between resonance K and topological channel weight.

Definition 2.6.1 (Topological Channel Weight)

For a topological channel with generalized Chern number K, define the channel weight:

$W[K] = \frac{1}{\Phi^K}$

- Larger K means higher fiber-bundle distortion, higher topological impedance, and smaller channel weight $W[K]$;
- Smaller K means larger channel weight.

Physical meaning: the channel weight $W[K]$characterizes the intrinsic ability of this topological pipe to transport the energy-particle flow. The higher the distortion (larger K), the higher the topological impedance and the lower the energy transmission ratio.

This is the mathematical source of Chapter 5's channel distribution ratios: corner ring $K = 10$, edge-midpoint ring $K = 5$, self-ring $K = 5$; all are substituted into this formula to compute the weights.

2.7 Chapter Summary

- Physical image: resonance K originates from the "five steps one circle" revolution-rotation nested phase unit. The standard integer first Chern number is applicable to compact closed bases; this theory's base space B is a non-compact layer axis, so the generalized first Chern number (resonance K) is introduced, allowing real values.
- Mathematical definition:

$$K \equiv \lim_{n^* \to n_{\max}} \frac{i}{2\pi} \int_{[0,n^*]} \Omega$$

- **K's dual identity (V4 core correction):** K is both the generalized first Chern number (topological distortion measure) and the resonance condition (phase-matching criterion). The two identities are not two parallel things, but two descriptions of the same quantity.
- Strictly distinguish two concepts:
    - n: base-space layer number (geometric layer number, coordinate parameter);
    - K: generalized first Chern number + resonance condition (topological distortion measure).
- Resonance condition: $r_{AB} = m/n$, $\Delta K = \log_{24}(m/n)$. Eigenfrequency: $\omega_K = \omega_0 \cdot 24^{-K}$.
- Shared K definition: two sets of fiber bundles are locally isomorphic, equivalently $K_A = K_B$ in the local region; it is the topological criterion for chemical-bond resonance matching and possesses topological stability.
- Hierarchical nesting of K: $K_1 \subset K_2 \subset \cdots \subset K_n$. Sharing K_n means sharing all lower-layer K's.
- V3's empirical splitting $\Delta K_{m+1} = 2.164 \Delta K_m$ is attributed to the higher-order contribution of the second generalized Chern number; the empirical numerical value is retained for computation, and the complete analytical derivation is left as an open problem.
- Define the topological channel weight $W[K] = 1/\Phi^K$, laying the groundwork for Chapter 5's four-force channel distribution.

One-sentence summary: Resonance K is the generalized first Chern number of a fiber bundle over a non-compact base space, describing the overall topological distortion of the spiral bundle, and simultaneously the resonance condition for phase matching; shared K means local fiber-bundle isomorphism; n is the coordinate layer number, K is the topological invariant and resonance condition, and the two are no longer confused.

Chapter 3 Mass Formula: Manifold Intersection and Area Overlap

3.0 Chapter Positioning

This chapter is the derivation layer and numerical verification layer of the entire book, and the core chapter of particle ontology.

In the V3 version, the masses of 33 particles were numerically fitted using elementary algebra and phenomenological formulas, achieving an overall average deviation of 0.89% and highly consistent results with experimental data, but lacking underlying geometric-topological explanation.

V4 goals for this chapter:

8. Physical image first: fountain image, lepton longitudinal spring, quark transverse spring, proton three-quark right-triangle geometric configuration;
9. Introduce differential topological transverse intersection and intersection number theory; particle rest mass is no longer an intrinsic input parameter but a topological invariant of submanifold transverse intersection;
10. Distinguish the manifold structures corresponding to leptons, quarks, gauge bosons, and composite hadrons (mesons, baryons); establish all mature V3 numerical results on the foundation of manifold intersection theory;
11. Establish the correspondence between the intersection number and Chapter 2's generalized first Chern number (resonance K), opening up the path between topological invariants and the mass spectrum;
12. Completely reproduce the 33-particle mass verification table, retaining all V3 experimental comparison data.

Inherited preliminary symbols: Φgolden ratio; $M_0 = 0.8255\ \mathrm{MeV}$baseline energy scale; K the generalized first Chern number (resonance K); $I(M_1, M_2)$submanifold intersection number.

Chapter 1-2 conclusions preliminary: the base-space layer number n is a coordinate parameter; K is the topological distortion invariant and resonance condition; the spiral is a fiber-bundle section; the particle is the topological jamming of the spiral wave on the Luoshu grid.

3.1 Physical Image First: Fountain, Longitudinal Spring, Transverse Spring

Before entering differential-topological mathematical tools, restore V3's physical intuition.

Imagine the energy-particle spiral flow gushing out from global Planck Center Point 5 (base space $n = 0$), i.e., the cosmic fountain image.

- **Leptons** (electron, muon, tau) correspond to **longitudinal springs**: the spiral section grows approximately longitudinally along the base-space layer axis, representing pure evolution along the fractal hierarchy; after topological jamming they form leptons.
- **Quarks** correspond to **transverse springs**: the spiral section extends in the transverse direction perpendicular to the layer axis, representing spatial-dimension expansion.

Taking the proton as the typical example: the proton is not a point-like object but a collective resonance of three transverse springs corresponding to three quarks. The three quark submanifolds form a right-triangle configuration in projection space, with three vertices corresponding to quark positions; rays are drawn from the global fractal center point to the three vertices, corresponding to the three sides of the 3-4-5 minimal integer right triangle. This right-orthogonal configuration is the geometric intuition of the yin-yang transverse energy balance under the strong interaction, and is also the image source of the subsequent proton mass geometric factor $24 = 2 \times (3 + 4 + 5)$.

Core proposition: The rest mass of a particle originates from the transverse intersection of submanifolds with each other; the degree of intersection is characterized by the intersection number; the intersection number couples with the generalized Chern number K, ultimately giving the mass spectrum.

Mass is topology; existence is interweaving.

3.2 Foundations of Manifold Intersection Theory

3.2.1 Transverse Intersection

Definition 3.2.1 (Transverse Intersection)

Let M_1 and M_2 be two submanifolds of an ambient smooth manifold M. If at every intersection point $p \in M_1 \cap M_2$, the two tangent spaces satisfy:

$$T_pM_1 + T_pM_2 = T_pM$$

then M_1 and M_2 are said to intersect transversely at point p.

Physical meaning of transverse intersection: the intersection point possesses topological stability. Small continuous deformations will not remove the intersection point; only a topological phase transition can destroy the intersection relation. This corresponds to the stability of elementary particles: small energy perturbations will not destroy the particle.

3.2.2 Intersection Number

Definition 3.2.2 (Intersection Number)

For two transversely intersecting oriented submanifolds M_1, M_2, the intersection number is defined as the algebraic sum of the signs of all intersection points:

$$I(M_1, M_2) = \sum_{p \in M_1 \cap M_2} \text{sign}(p)$$

The intersection number I is an integer topological invariant, independent of the metric, describing only the topological number of times the submanifolds interpenetrate and interweave.

Core hypothesis of this theory: the rest mass of a particle is proportional to the intersection number of the corresponding internal submanifolds.

$$M \propto I \times M_0$$

$M_0 = 0.8255\ \mathrm{MeV}$ is the baseline zero-point energy scale of this theory, corresponding to the projection of vacuum zero-point energy onto the fractal fiber bundle.

3.3 Lepton Masses: Self-Intersection of Fiber-Bundle Sections

3.3.1 Physical Model

Leptons such as the electron, muon, and tau correspond to a single spiral section $s_e : B \to E$ of the fiber bundle.

Lepton mass originates from the self-intersection of the spiral section: the same spiral section undergoes submanifold self-intersection; the self-intersection number $I(s, s)$ is modulated by the generalized Chern number K of that particle.

Note: self-intersection is not the section intersecting on the base space B; rather, self-intersection occurs within the total-space fiber bundle E, and its projection onto the base space appears only as the "topological jamming" standing-wave layer.

General lepton mass formula:

$$M_l = M_0 \cdot \Phi^{k_l}$$

- $M_0 = 0.8255\ \mathrm{MeV}$ baseline energy;
- k_l is the golden exponent after the self-intersection number mapping, in one-to-one correspondence with the generalized Chern number K_l of that lepton;
- The standing-wave locking layer number n_l ($n_e = 16.79$, $n_\mu = 14.55$, $n_\tau = 13.64$) is the position where self-intersection-topological jamming occurs, not the topological origin.

3.3.2 Electron

Electron: $k_e = 1$

$$M_e = M_0 \cdot \Phi^1 = 0.8255 \times 0.618034 = 0.5102\ \mathrm{MeV}$$

Experimental value 0.511 MeV; relative deviation -0.16%.

3.3.3 Muon

Muon: $K_e - K_\mu = 2.24$, $\delta = 0$, $k_\mu = 1 - 5 \times 2.24 = -10.2$.

$$M_\mu = M_0 \cdot \Phi^{-10.2} = 111.9\ \mathrm{MeV}$$

Experimental value 105.7 MeV; relative deviation $+5.85\%$.

3.3.4 Tau

Tau: $K_e - K_\tau = 3.15 > 3$, deep-layer locking triggered, $\delta = 2\Phi = 1.236$, $k_\tau = 1 - 5 \times 3.15 - 1.236 = -15.986$.

$$M_\tau = M_0 \cdot \Phi^{-15.986} = 1808 \text{ MeV}$$

Experimental value 1777 MeV; relative deviation $+1.75\%$.

3.3.5 Lepton Summary Table

Particle	Theoretical value (MeV)	Experimental value (MeV)	Relative deviation
Electron	0.5102	0.511	鉴?.16%
Muon	111.9	105.7	+5.85%
Tau	1808	1777	+1.75%

3.4 Quark Masses: Multi-Submanifold Direct-Sum Intersection

Quarks correspond to transversely extending submanifolds; according to Chapter 2's generalized Chern number direct-sum property $K(E_1 \oplus E_2) = K(E_1) + K(E_2)$, the intersection number of quark manifolds is obtained by superposition of direct-sum submanifolds.

General quark mass form:

$$M_q = M_{\text{base}} \cdot 24^{I_q}$$

M_{base} is the corresponding family baseline mass; I_q is the quark intersection index, coming from the topological counting of submanifold transverse intersection.

3.4.1 Six-Flavor Quark Numerical Results

Quark flavor	Intersection index I_q	Theoretical value (MeV)	Experimental value (MeV)	Deviation
u (up)	0.4691	2.27	2.30	鉴?.30%
d (down)	0.6976	4.69	4.80	鉴?.20%
s (strange)	鉴?.0309	95.80	95.00	+0.80%
c (charm)	0.7809	1265.00	1275.00	鉴?.40%
b (bottom)	0.2691	4179.60	4180.00	鉴?.01%

Quark flavor	Intersection index I_q	Theoretical value (MeV)	Experimental value (MeV)	Deviation
t (top)	1.4451	175600	172760	+1.50%

Note: The six-flavor quark intersection indices I_q form a Fibonacci-like sequence, embodying the self-similarity of the fractal in the flavor degrees of freedom. There exists a one-to-one mapping between the intersection index and the generalized Chern number K_q; the complete analytical derivation of this mapping is left as an open problem.

3.5 Proton Mass: Three-Quark Submanifold Transverse Intersection

The proton is composed of u, u, d three quark transverse submanifolds, which form a 3-4-5 right-triangle configuration in projection space, and the submanifolds undergo triple transverse intersection in the neighborhood of the fractal local center point.

The intersection number alone gives the topological cardinality; a set of geometric correction factors is also needed, coming from fractal projection, spatial convergence, and QED outer-skin radiation correction.

$$M_p = M_e \times 24^2 \times \pi \times \left(1 + \tfrac{2}{137}\right)$$

- $24^2 : 24 = 2 \times (3 + 4 + 5)$, coming from the edge-length sum of the left- and right-handed mirror images of the 3-4-5 right triangle, corresponding to the geometric factors of triple spatial projection and energy convergence;
- π: the correction of the outward-radiation projection equivalent circular area of the right-triangle configuration under the high-energy limit of the strong interaction;
- $1 + \frac{2}{137}$: the electromagnetic outer-skin radiation correction brought by the fine-structure constant.

Substituting numerical values:

$$M_p = 0.511 \times 576 \times 3.14159 \times 1.0146 = 938.27\ \mathrm{MeV}$$

Experimental value 938.27 MeV, deviation about 0%.

3.5.1 Origin of $2/137$

The proton outer skin has two faces:

- **Inner surface**: facing the proton interior, the interface between quarks and corner points;
- **Outer surface**: facing the yang-side light sea, the interface between the proton and the light sea.

Each face contributes one fine-structure constant $\alpha = 1/137$. Therefore, the coefficient is $2/137$.

Outer-skin energy:

$$E_{\text{outer skin}} = M_{\text{bare}} \times \frac{2}{137} = 924.7 \times 0.0146 = 13.5\ \text{MeV}$$

This is the origin of the 13.6 MeV in the proton mass formula.

Physical interpretation: The proton is not simply the sum of three quark masses; the vast majority of the proton mass comes from the topological emergent contribution brought by submanifold intersection, comparable to the modern QCD phenomenon that "most of the mass comes from gluon-field interactions." In this theory, gluons correspond to the umbilical-cord connection between the three quarks.

3.6 Composite Particles (Mesons, Baryons): Submanifold Intersection Superposition

Mesons are pairwise transverse intersections of quark-antiquark submanifolds; baryons are triple intersections of three-quark submanifolds.

General formula:

$$M_{\text{comp}} = \sum M_q + E_{\text{bind}}$$

$\sum M_q$is the sum of the constituent quark masses; $E_{\text{bind}}$is the topological binding energy given by the intersection overlap area. The binding energy is determined by the intersection overlap ratio of the submanifolds.

3.6.1 Representative Composite Particle Verification

Particle	Constituents	Theoretical value (MeV)	Experimental value (MeV)	Deviation
π^+	u+d	140.96	139.57	<2%
K^+	u+s	500.1	493.68	+1.3%
J/ψ	c+c	3097	3096.9	鋸?%
Υ	b+b	9467.2	9460.3	鋸?%
n (neutron)	u+d+d	939.60	939.57	鋸?%
Ω^-	s+s+s	1672.4	1672.5	鋸?%

3.7 Gauge Boson Masses: Fiber-Bundle Curvature Integral Contribution

The masses of W, Z, and Higgs bosons originate from the integral contribution of the fiber-bundle curvature $\Omega = d\omega$.

$M = M_b \cdot \Phi^{-5} \cdot f$

$M_b = 4179.6\ \mathrm{MeV}$ is the bottom-quark baseline mass; $\Phi^{-5} \approx 11.09$; f is the corresponding gauge-channel geometric factor.

Particle	Geometric factor f	Theoretical value (GeV)	Experimental value (GeV)	Deviation
W boson	$\sqrt{3}$	80.3	80.4	鋆?.1%
Z boson	$\sqrt{3} \times \sqrt{107/83}$	91.1	91.2	鋆?.1%
Higgs	$\sqrt{3} \times \pi/2$	126.1	125.1	+0.7%

3.8 Overall Statistical Summary of 33 Particles

Category	Number	Average relative deviation	Maximum deviation
Leptons	3	2.59%	5.85% (muon)
Quarks	6	0.76%	1.50% (t quark)
Gauge bosons	3	0.28%	0.70% (Higgs)
Composite particles	21	0.52%	1.90%
Total	33	0.89%	5.85%

This statistic inherits all V3 numerical results; V4's contribution is to provide a manifold-intersection 鈥慽ber-bundle-topological geometric explanation for this set of numerical results.

3.9 Chapter Summary

- Physical image: leptons correspond to longitudinal spiral springs, quarks to transverse spiral springs; the proton's three quarks form a 3-4-5 right-triangle orthogonal configuration.
- Particle rest mass is a topological invariant of submanifold transverse intersection, $M \propto I \cdot M_0$.
- Leptons come from fiber-bundle section self-intersection; quarks from submanifold direct-sum intersection; gauge bosons from curvature integrals; composite particles

from superposition of constituent intersections, with mass equal to the sum of quark masses plus topological binding energy.

- Proton mass geometric correction factors: 24^2 comes from the left- and right-handed mirror images of the 3-4-5 right triangle; π comes from the equivalent-circular-area correction of the triangle configuration's outward-radiation projection; $2/137$ comes from the two faces of the outer skin each contributing one fine-structure constant.
- Completely reproduce the 33-particle mass verification, with overall average deviation 0.89%; all V3 experimental comparison results are inherited.
- Establish the correspondence between the intersection number and Chapter 2's generalized first Chern number (resonance K); open up the path between particle ontological mass and topological invariants.

One-sentence summary: The mass of all things originates from the transverse intersection of fractal submanifolds; the intersection number and the generalized Chern number K jointly determine the particle mass spectrum. The proton's three quarks intersect orthogonally in a 3-4-5 right-triangle configuration; $24 = 2 \times (3 + 4 + 5)$ is precisely the number-theoretic expression of this right-angle geometry.

Chapter 4 24-Fold Amplification: Symplectic Geometry and Hamiltonian Flow

4.0 Chapter Positioning

This chapter is the core dynamical derivation layer of the entire book.

V3 Core Background Review:

In V3, the 24-fold amplification factor is the core empirical constant of the entire fractal system: each increase of the base-space layer number n by 1 amplifies the characteristic spatial scale of the Luoshu grid by 24 times. V3 provided two independent geometric sources:

1. Source ①: The 3-4-5 minimal integer right triangle, left- and right-handed mirror images, $2 \times (3 + 4 + 5) = 24$;
2. Source ②: Five-steps-one-circle revolution-rotation phase counting: revolution 5 steps, each step with 5 subdivided phases, total phase units $5 \times 5 = 25$, starting and ending points topologically coincident, subtracting 1, giving the number of independent phases $25 - 1 = 24$.

The key gap left by V3: V3 only provided the image and counting results, and did not answer a key contradiction: continuous Hamiltonian flow undergoing exponential expansion gives the natural constant $e \approx 2.71828$; why is this theory's fractal amplification factor 24? A gap exists

between the continuous solution of continuous symplectic geometry and the discrete Luoshu grid observational results, and V3 did not mathematically clarify the relationship between "continuous dynamics" and the "discrete fractal grid."

V4 Core Tasks for This Chapter:

1. Physical image first, completely inherit V3's two original image sets: fountain rotation-revolution, five-steps-one-circle phase folding, 3-4-5 right-triangle left-right chirality;
2. Systematically introduce the full mathematical toolbox of symplectic manifolds, symplectic forms, Hamiltonian vector fields, and Hamiltonian flows;
3. Establish the Hamiltonian function of this theory's fractal system, solve the Hamiltonian flow, obtain the continuous solution, and derive the continuous expansion factor e;
4. Strictly distinguish: e is the expansion factor of the continuous phase space; 24 is the physical amplification factor obtained after the continuous symplectic flow is sampled/discretized on the Luoshu discrete grid;
5. Completely reproduce the two independent paths deriving 24, translating all of V3's images into the symplectic-geometric framework;
6. Connect with Chapter 1's base-space metric $ds_B^2 = (\ln 24)^2 dn^2$, explaining the dynamical root of $\ln 24$ in the metric.

Inherited preliminary symbols: Φ golden ratio; base space B layer axis; S^1 phase circle fiber; generalized Chern number K; 3-4-5 right triangle.

4.1 Physical Image First: Fountain's Rotation, Revolution, and Phase Folding

Before entering the abstract mathematical definitions of symplectic geometry, completely inherit V3's physical image.

Imagine the cosmic fountain ejected outward by global Planck Center Point 5 ($n = 0$). Spiral motion simultaneously contains two motion modes: revolution and rotation.

- **Revolution**: the spiral advances outward along the base-space layer axis, following V3's "five steps one circle" law: base-space layer advances $\Delta n = 5$, the phase rotates a full 2π, with each revolution step contributing a phase increment of 72°;
- **Rotation**: inside each revolution step, the spiral itself also has high-frequency subdivided rotation; each step is further subdivided into 5 rotation phase units, with each subdivided phase rotating $72^\circ/5 = 14.4^\circ$.

Full phase-unit counting:

Revolution 5 steps × 5 rotation subdivided phases per step = 25 phase positions.

Because the spiral is a closed periodic orbit, the 25th phase position and the initial 1st phase position are topologically coincident, belonging to the same physical state, and the duplicate count must be subtracted.

$$N_{\text{independent phases}} = 5 \times 5 - 1 = 24$$

This is 24 in the sense of phase counting.

The second image source inherits V3's 3-4-5 minimal integer right triangle:

The 3-4-5 right triangle is the smallest right triangle with all integer side lengths, with perimeter $L = 3 + 4 + 5 = 12$. The fractal spiral has left-handed and right-handed mirror-image sets; superposing the two, we obtain $2 \times (3 + 4 + 5) = 24$.

Important image proposition: 24 is not an artificial fitting parameter; it has two independent image sources: ① revolution-rotation phase subtracting coincidence; ② 3-4-5 right triangle left-right chirality mirror images.

But both of the above belong to discrete-level geometric counting; we need to answer: why does the continuous dynamics converge to this discrete value — this is the problem that symplectic geometry and Hamiltonian flow must solve.

Important distinction:

- **Continuous symplectic flow**: describes ideal dynamics of the phase space with no grid constraint, giving the expansion factor e;
- **Luoshu grid**: a discrete topological constraint that samples the continuous Hamiltonian flow, outputting the physical observable amplification factor 24.

4.2 Foundations of Symplectic Geometry

4.2.1 Symplectic Manifold

Definition 4.2.1 (Symplectic Manifold)

An even-dimensional smooth manifold M equipped with a closed, non-degenerate 2-form ω, called the symplectic form, is called a symplectic manifold (M, ω).

Two core axioms:

1. **Closure condition**: $d\omega = 0$; the exterior derivative of the symplectic form vanishes, corresponding to conservation of phase-space volume, i.e., the Liouville theorem of classical mechanics;
2. **Non-degeneracy**: for any nonzero tangent vector X, there does not exist $\omega(X, \cdot) = 0$ identically; this ensures a one-to-one correspondence between vector fields and 1-forms via ω.

Physical meaning: the symplectic manifold is the phase space of Hamiltonian mechanics, with coordinates containing position (in this theory corresponding to the layer number n) and phase (in this theory corresponding to the spiral phase θ).

4.2.2 Hamiltonian Vector Field

Definition 4.2.2 (Hamiltonian Vector Field)

Given a smooth Hamiltonian function $H : M \to \mathbb{R}$, the Hamiltonian vector field X_H is the unique vector field satisfying:

$$\omega(X_H, \cdot) = dH$$

- dH is the exterior derivative of the Hamiltonian function (energy-gradient 1-form);
- the symplectic form ω acts as a "converter": mapping the energy gradient to the direction-of-motion vector X_H in phase space.

4.2.3 Hamiltonian Flow

Definition 4.2.3 (Hamiltonian Flow)

The one-parameter diffeomorphism group $\phi_t : M \to M$ generated by the Hamiltonian vector field X_H is called the Hamiltonian flow.

ϕ_t describes the evolution of a point in phase space with time t, satisfying

$$\frac{d}{dt}\phi_t = X_H \circ \phi_t$$

Liouville theorem corollary: the Hamiltonian flow preserves the symplectic form $\phi_t^* \omega = \omega$; phase-space volume is strictly conserved during evolution.

4.3 Hamiltonian Function and Hamiltonian Flow of This Theory's Fractal System

4.3.1 Construction of the Hamiltonian Function

Inheriting Chapter 1 and Chapter 2's spiral section, where the spiral phase evolution is determined by the connection coefficient $A_n = \ln(1/\Phi^5)$, take the Hamiltonian function:

$$H(n, \theta) = \Phi^5 \cdot n$$

- n: base-space layer number (phase-space position coordinate);
- θ: spiral phase (phase-space angular coordinate);
- $\Phi^5 \approx 0.09017$, coming from the V3-V4 umbilical-cord longitudinal-channel decay constant.

Physical interpretation: The Hamiltonian H represents the phase-space energy of the fractal spiral system; the energy is proportional to the layer number n.

On this theory's phase space (n, θ), the symplectic form is taken as the standard polar-coordinate symplectic form:

$$\omega = dn \wedge d\theta$$

4.3.2 Solving the Hamiltonian Vector Field

$$dH = \frac{\partial H}{\partial n} dn + \frac{\partial H}{\partial \theta} d\theta = \Phi^5 dn$$

From $\omega(X_H, \cdot) = dH$, substituting $\omega = dn \wedge d\theta$, we obtain the Hamiltonian vector field:

$$X_H = -\Phi^5 \frac{\partial}{\partial \theta}$$

4.3.3 Hamiltonian Flow Differential Equation System

The vector field directly gives the evolution equation system:

$$\frac{dn}{dt} = 0$$

$$\frac{d\theta}{dt} = -\Phi^5$$

Physical interpretation:

1. $\frac{dn}{dt} = 0$: within the Hamiltonian flow evolution period, the layer number n remains unchanged;
2. $\frac{d\theta}{dt} = -\Phi^5$: only the phase θrotates at a constant angular velocity.

4.3.4 Periodic Orbit: Five Steps One Circle

Rotating a full circle $\Delta\theta = 2\pi$, solve for the period T:

$$T = \frac{2\pi}{\Phi^5}$$

Substituting $\Phi^5 \approx 0.09017$:

$$T = \frac{2\pi}{0.09017} \approx 69.68$$

T is the characteristic evolution period of one spiral revolution.

Proposition: $\theta(t + T) = \theta(t) + 2\pi$; this is the Hamiltonian periodic orbit corresponding to V3's "five steps one circle." At the continuous dynamics level, after each characteristic period T the phase completes a full revolution.

4.4 Continuous Expansion: Why the Continuous Solution Gives e, Not 24

Now study the expansion motion of the spiral radius with time. The spiral characteristic radius $r(t)$satisfies the differential equation (continuous no-grid assumption):

$$\frac{dr}{dt} = \frac{r}{T}$$

Physical meaning: the relative expansion rate of the radius equals the reciprocal of the period. This is a first-order ordinary differential equation, with initial condition $t = 0, r = r_0$.

$$r(t) = r_0 e^{t/T}$$

After one full revolution period $t = T$, the radius amplification factor:

$$\frac{r(T)}{r_0} = e^{T/T} = e \approx 2.71828$$

Key conclusion: In the continuous symplectic phase space without discrete grid constraints, each revolution of the spiral gives a radius amplification factor of the natural constant e, not 24.

This is the core contradiction that V3-V4 needs to solve: continuous dynamics gives e; the physical fractal universe observation gives 24. The two do not conflict; they belong to different physical premises.

4.5 Luoshu Grid: Discretized Sampling of the Symplectic Manifold

4.5.1 Distinguishing the Physical Premises of Continuous vs Discrete

- **Continuous symplectic flow (mathematical ideal model)**: phase space M is completely smooth, with no minimum-scale grid; the expansion factor e is a purely mathematical continuous-dynamics result;
- **Physical universe (Luoshu fractal grid)**: the real fractal universe is not an infinitely smooth continuous manifold; there exists an underlying discrete Luoshu grid structure; the continuous Hamiltonian flow motion is subjected to discrete sampling constraints by this grid.

V4 proposition: The 24-fold amplification is the physical observable amplification factor output after the continuous Hamiltonian periodic orbit is sampled on the Luoshu discrete grid.

Physical image restatement: In the continuous phase space, the spiral can take any phase position; but the Luoshu grid only allows the spiral to land on 24 independent phase lattice points. The continuously moving spiral can only dock at discrete allowed lattice points; each complete revolution it jumps to the next-level lattice point, corresponding to a 24-fold spatial-scale amplification.

Returning to 4.1's phase counting: $5 \times 5 - 1 = 24$; these 24 lattice points are precisely the set of independent phase lattice points allowed by the Luoshu grid. The continuous flow is the underlying dynamics; 24 is the observable physical result after the grid imposes topological constraints.

4.5.2 Two Independent Paths Deriving 24

Path one [phase counting path]: revolution 5 steps, each step with 5 rotation phases, subtracting the topological coincidence 1, $24 = 5 \times 5 - 1$, coming from the discrete counting of spiral phase units;

Path two [geometric number-theoretic path]: 3-4-5 minimal integer right triangle, left-right chirality mirror images, $24 = 2 \times (3 + 4 + 5)$.

V4 interpretation:

- Path one comes from the discrete phase division of the fiber-bundle phase circle S^1;
- Path two comes from the metric geometry of this theory's underlying fractal space.

Two completely independent lines of thought arrive at the same constant 24; this is not a numerical coincidence but a self-consistency requirement of this theory's system.

4.5.3 Connection with Chapter 1's Base-Space Metric

Chapter 1's base-space metric:

$$ds_B^2 = (\ln 24)^2 dn^2$$

Now we can provide the dynamical explanation:

Each increase of the layer number by $dn = 1$ amplifies the true spatial characteristic scale by 24 times; the scale transformation factor $\propto 24^{dn}$; taking the differential gives $ds_B \propto \ln 24 \, dn$.

$\ln 24$ is not an artificially input parameter; it comes from the discrete-grid sampling output amplification factor 24.

Note: $\ln 24$ does not come from the continuous expansion factor $\ln e = 1$; it comes from the discrete sampling output 24. This again distinguishes continuous e from discrete 24.

4.6 Physical Corollaries of the Constant 24

1. The scale amplification of this theory's fractal universe is not directly output by continuous dynamics; the continuous symplectic flow gives e, and the Luoshu discrete grid sampling outputs the physical amplification factor 24;
2. $24 = 5 \times 5 - 1$: the phase- spiral periodic angle source;
3. $24 = 2 \times (3 + 4 + 5)$: 3-4-5 right triangle left-right chirality mirror images, corresponding to the proton three-quark right-triangle configuration (Chapter 3); opening up the path between Chapter 3's particle geometry and Chapter 4's fractal amplification constant;
4. The base-space metric coefficient $\ln 24$ is determined by the discrete sampling result;
5. Chapter 1's spiral characteristic radius $r_n = \frac{\ell_P \cdot 24^n}{2\pi}$, with base number 24 being precisely this chapter's discrete sampling result.

4.7 Chapter Summary

1. Physical image inherits V3: the fountain spiral simultaneously has revolution and rotation; the five-steps-one-circle periodic orbit; 24 has two independent sources: ① $5 \times 5 - 1 = 24$phase counting; ② 3-4-5 right triangle left-right chirality mirror images $2 \times (3 + 4 + 5) = 24$.
2. Symplectic geometry foundations: symplectic manifold, symplectic form, Hamiltonian vector field, Hamiltonian flow; Liouville theorem guarantees phase-space volume conservation.
3. Construct this theory's Hamiltonian function $H(n, \theta) = \Phi^5 \cdot n$, solve the Hamiltonian flow, obtain the revolution period $T = 2\pi/\Phi^5 \approx 69.68$, strictly deriving V3's "five steps one circle" as a periodic orbit.
4. Core distinction: the no-grid continuous symplectic flow gives an expansion factor per revolution of e; the physical universe has the Luoshu discrete grid, the continuous flow is sampled by the grid, outputting the physical observable amplification factor 24. The premises of e and 24 are different; there is no contradiction.
5. The coefficient $\ln 24$in the base-space metric $ds_B^2 = (\ln 24)^2 dn^2$ comes from the discrete sampling output 24; opening up the path between Chapter 1's fiber-bundle geometry and this chapter's symplectic-geometric dynamics.
6. Opening up the connection with Chapter 3: the 3-4-5 triangle source of 24 corresponds to the proton three-quark 3-4-5 right-triangle orthogonal intersection configuration.

One-sentence summary: The continuous Hamiltonian flow gives e; after sampling on the Luoshu discrete grid it gives the 24-fold fractal amplification factor; 24 simultaneously comes from the phase folding count and the 3-4-5 right-triangle left-right chirality mirror images, two independent paths.

Chapter 5 Channel Distribution Ratios: Topological Invariants and Shadow-Dynamics Prerequisites

5.0 Chapter Positioning

This chapter belongs to the derivation layer- topological layer and is simultaneously the preliminary foundational chapter for shadow dynamics.

Review of V3's original work: V3 obtained the energy weights of three topological channel sets through elementary algebra and calculated the energy distribution ratios of the strong force, electromagnetic interaction, and gravity: 89.87%, 8.10%, 2.03%. But V3 did not establish the

weights on fiber-bundle topological invariants; the weight formula was a phenomenological algebraic result.

V4 Core Tasks for This Chapter:

- Image first, completely inherit V3's Luoshu three orthogonal ring physical images: corner ring, vertical edge-midpoint ring, horizontal edge-midpoint ring, gravity self-ring; clarify the physical meaning of topological stacking;
- Relying on Chapter 2's generalized first Chern number (resonance K), strictly define the topological channel weight $W[K] = 1/\Phi^K$;
- Give respectively the Chern-number sources of the corner ring, edge-midpoint ring, and gravity self-ring: corner ring $K = 10$, edge-midpoint ring $K = 5$, gravity self-ring $K = 5$;
- Complete the total weight calculation, reproducing V3's three sets of distribution ratios 89.87%, 8.10%, 2.03%;
- Introduce the 9th-order light matrix concept, eigenvalue multiplicities, and response functions, establishing the mapping from topological channels to observable coupling constants;
- Build the interface leading to Chapter 7's topologically upgraded shadow dynamics, giving the preliminary form of the shadow-dynamics general formula.

Inherited preliminary symbols:

Φgolden ratio; $\Phi^5 \approx 0.09017$, $1/\Phi^5 \approx 11.090$, $1/\Phi^{10} \approx 122.991$;

K: generalized first Chern number; $W[K] = 1/\Phi^K$ topological channel weight;

Three orthogonal ring sets: corner ring $2 \to 4 \to 8 \to 6 \to 2$; vertical and horizontal edge-midpoint rings $1 \to 3 \to 9 \to 7 \to 1$; gravity self-ring.

5.1 Physical Image First: Topological Stacking of the Three Orthogonal Rings

Returning to the Luoshu nine-grid three-dimensional skeleton image. From global Planck Center Point 5 (base space $n = 0$), three mutually orthogonal vortex ring pipe sets extend outward, i.e., umbilical-cord vortex tubes.

- **Corner ring**: circulation sequence $2 \to 4 \to 8 \to 6 \to 2$, corresponding to the strong-interaction channel. Corner points represent matter poles, each with equivalent step number $2 \times 5 = 10$. The corner ring has 4 independent topological pipes.
- **Edge-midpoint ring**: divided into vertical edge-midpoint ring and horizontal edge-midpoint ring, circulation sequence $1 \to 3 \to 9 \to 7 \to 1$, corresponding to the electromagnetic-interaction channel. The edge-midpoint equivalent step number is $1 \times 5 = 5$. The edge-midpoint rings have 4 independent topological pipes in total.
- **Gravity self-ring**: longitudinal- transverse coupled self-closed topological channel, corresponding to the gravitational interaction, with only 1 independent pipe, equivalent step number $1 \times 5 = 5$.

Physical meaning of topological stacking:

After projection, the vertical edge-midpoint ring and horizontal edge-midpoint ring land on the same set of grid nodes $\{1,3,9,7\}$, undergoing topological stacking; they share grid nodes but belong to two orthogonal vortex tubes, physically independent. The corner ring projects onto the corner nodes $\{2,4,6,8\}$, forming an independent subspace.

Important distinction: stacking is not coincidence. Stacking refers to different sub-bundles of the higher-dimensional fiber bundle projecting onto the same node in the lower-dimensional Luoshu grid, while the vortex tubes themselves are mutually distinguished in the higher-dimensional total space.

Physical interpretation: It is precisely this asymmetry of the topological structure that directly causes the enormous differences in the energy ratios of the four fundamental forces.

5.2 Definition of Topological Channel Weight (Based on Generalized Chern Number K)

Definition 5.2.1 (Topological Channel Weight)

For a given topological pipe with generalized first Chern number K, the topological weight of that pipe is:

$$W[K] = \frac{1}{\Phi^K}$$

- Larger K means higher fiber-bundle distortion, greater topological impedance, and smaller channel weight $W[K]$;
- Smaller K means lower distortion, smaller impedance, and larger channel weight.

Physical image: the weight $W[K]$characterizes the intrinsic ability of this vortex tube pipe to transport the energy-particle flow; it is a purely topological invariant, independent of the metric and of external conditions.

5.2.1 Generalized Chern Number Values of Each Ring

Corner ring (strong-interaction channel): Corner points are matter poles, requiring double phase cycles to complete closure; equivalent step number $2 \times 5 = 10$, therefore corner ring $K_{\text{corner}} = 10$.

$$W_{\text{corner}} = W[10] = \frac{1}{\Phi^{10}} \approx 122.991$$

$$W_{\text{corner total}} = 4 \cdot W_{\text{corner}} = 4 \times 122.991 = 491.964$$

Edge-midpoint ring (electromagnetic-interaction channel): Edge-midpoints are at the equatorial position, requiring single phase cycles for closure; equivalent step number $1 \times 5 = 5$, $K_{\text{edge}} = 5$.

$$W_{\text{edge}} = W[5] = \frac{1}{\Phi^5} \approx 11.090$$

$$W_{\text{edge total}} = 4 \cdot W_{\text{edge}} = 4 \times 11.090 = 44.360$$

Gravity self-ring (gravity channel): Self-closed transverse topological channel, $K_{\text{self}} = 5$, only 1 pipe.

$$W_{\text{self total}} = 1 \cdot W[5] = 11.090$$

5.2.2 System Total Weight

$$W_{\text{total}} = W_{\text{corner total}} + W_{\text{edge total}} + W_{\text{self total}} = 491.964 + 44.360 + 11.090 = 547.414$$

5.2.3 Topological Distribution Ratios of the Three Forces

$$\eta_{\text{strong}} = \frac{W_{\text{corner total}}}{W_{\text{total}}} = \frac{491.964}{547.414} \approx 89.87\%$$

$$\eta_{\text{electromagnetic}} = \frac{W_{\text{edge total}}}{W_{\text{total}}} = \frac{44.360}{547.414} \approx 8.10\%$$

$$\eta_{\text{gravity}} = \frac{W_{\text{self total}}}{W_{\text{total}}} = \frac{11.090}{547.414} \approx 2.03\%$$

Result: Completely reproduce V3's three sets of energy distribution ratios. V4's improvement: no longer pure algebraic fitting; the ratios all come from the topological weights determined by the fiber-bundle generalized Chern number K.

Important note: This is the topological intrinsic ratio, not yet including response-function corrections; the topological ratio is a property of the pipe itself; the observable coupling constant must additionally multiply the response function of the corresponding channel.

5.3 Light Matrix: 9×9 Hermitian Matrix, Eigenvalue Multiplicities

V3 proposed the light matrix concept, used to map the topological pipe weights to the observable coupling constants of the four fundamental interactions. V4 places it after the topological weights.

Definition 5.3.1 (Light Matrix)

The light matrix L is a 9×9 Hermitian matrix, corresponding to all 9 degrees of freedom of the Luoshu grid. The matrix's 9 eigenvalues are distributed to the four interaction classes according to eigenvalue multiplicity:

- Gravity mode: eigenvalue multiplicity = 1; corresponding to the single pipe of the gravity self-ring;
- Strong mode: eigenvalue multiplicity = 4; corresponding to the 4 corner-ring pipes;
- Electromagnetic mode: eigenvalue multiplicity = 2; corresponding to the orthogonal polarizations of the vertical and horizontal edge-midpoint rings;
- Weak mode: eigenvalue multiplicity = 2; corresponding to the two chiral channels of parity non-conservation.

Sum of all multiplicities: $1 + 4 + 2 + 2 = 9$, exactly equal to the 9 degrees of freedom of the Luoshu nine-grid.

5.3.2 Response Functions

Each interaction class corresponds to a response function R; the response function describes the actual conduction efficiency of the energy-particle flow in the topological vortex tube under the real cosmic-web environment.

$$R_{\text{gravity}} = 1$$

$$R_{\text{strong}} = 1.485$$

$$R_{\text{electromagnetic}} = 1.626$$

$$R_{\text{weak}} = 1.25$$

Physical interpretation: The topological weight $W[K]$is the intrinsic topological ability of the pipe; the response function R represents the correction brought by the external cosmic-web environment, vortex-tube coupling, and vacuum medium. The observable coupling constant is proportional to $W[K] \cdot R$.

The analytical derivation of the response function from the first principles of fiber bundles and the Helmholtz- Moffatt vortex theory is left as a subsequent open research topic.

5.4 Interface to Shadow Dynamics

The topological weights, ratios, and response functions obtained in this chapter are the core input parameters for the topologically upgraded shadow dynamics.

General force formula of shadow dynamics (preliminary definition, complete numerical calculation placed in Chapter 7):

$$F = P_0^{(\text{local})} \cdot \eta \cdot W[K] \cdot R \cdot A_{\text{eff}}$$

- $P_0^{(\text{local})}$: local cosmic-web pressure, coming from the global Planck source + stellar local generator superposition (Chapter 12 cosmology);
- η: shadow interception rate, geometric obstruction factor;
- $W[K]$: this chapter's topological channel weight;
- R: channel response function;
- A_{eff} : effective geometric cross-section, coming from Chapter 3's manifold-intersection theory particle submanifold projection.

Connecting explanation: This formula opens up multiple main lines of the entire book:

- Particle ontology (Chapter 3 manifold intersection) outputs A_{eff};
- This chapter's pipe topology gives $W[K], R$;

- Cosmology (Chapter 12 stellar generator) gives $P_0^{(\text{local})}$;
- Chapter 7 substitutes into this formula, completing respectively the numerical verification of gravity, Coulomb (electromagnetic force), and the strong force.

Key re-distinction (inheriting Chapters 1 and 7 important corrections): The longitudinal umbilical-cord channel has Φ^5 decay; the transverse closed self-ring channel corresponding to gravity obeys the holonomy condition $\text{Hol}(\gamma) = 1$ and does not have exponential decay. The weight $W[K = 5]$ only represents topological impedance, not cross-layer exponential decay. V3's old gravitational formula misusing the longitudinal umbilical-cord decay factor is discarded.

5.5 Chapter Summary

- Physical image: three orthogonal vortex pipe sets: corner ring (strong force $K = 10$), edge-midpoint ring (electromagnetism $K = 5$), gravity self-ring (gravity $K = 5$); distinguish higher-dimensional topological stacking from lower-dimensional grid projection coincidence.
- Relying on Chapter 2's generalized first Chern number K, define the topological channel weight $W[K] = 1/\Phi^K$; the weight is a topological invariant, characterizing the intrinsic ability of the vortex tube to transport the energy-substance flow.
- Substituting the number of pipes of each ring, the total weight $W_{\text{total}} = 547.414$ is calculated, reproducing V3's topological ratios: strong force 89.87%, electromagnetic force 8.10%, gravity 2.03%.
- Introduce the 9th-order Hermitian light matrix; the 9 eigenvalue multiplicities are strictly distributed to the four interaction classes, satisfying $1 + 4 + 2 + 2 = 9$ Luoshu degree-of-freedom completeness; introduce the response function R as the conduction-efficiency correction of the environmental medium to the topological pipes.
- Give the preliminary form of the topologically upgraded shadow-dynamics general formula; establish the interface leading to Chapter 7's gravity-Coulomb force numerical calculation; the effective cross-section A_{eff} comes from Chapter 3's manifold intersection, and the local pressure $P_0^{(\text{local})}$ comes from Chapter 12's stellar generator cosmology.
- Important correction restatement: gravity's transverse closed channel $\text{Hol}(\gamma) = 1$; discard V3's gravity old longitudinal umbilical-cord exponential decay; the longitudinal umbilical-cord decay is only used for the particle-generation longitudinal channel.

One-sentence summary: The energy distribution ratios of the four fundamental forces come from the topological weights determined by the generalized Chern numbers K of different topological vortex tubes; the light-matrix eigenvalue multiplicities complete the 9-degree-of-freedom distribution; this chapter builds all the topological inputs for shadow dynamics.

Chapter 6 Ionization Energy: Double-Vortex Intersection and Breathing Correction

6.0 Chapter Positioning

This chapter is the derivation layer and numerical verification layer, and the core chapter of the atomic structure section.

In V3, the ionization energy calculation adopted an interpolation method, achieving good agreement with experiment, but the interpolation parameters carried a certain empirical component. This chapter uses the first-principles re-derivation via the "double-vortex intersection" model, retaining V3's final numerical results and reducing the interpolation terms back to geometric quantities.

Core tasks of this chapter:

1. Image first, introducing the "double-vortex intersection" model, reinterpreting the electron orbit as the phase-matching surface between the Planck vortex and the atomic-nucleus vortex.
2. Establish the relative depth coordinate system, with the atomic nucleus as origin, deriving the effective depth of each electron orbit.
3. Strictly derive the phase-matching condition $\Delta\phi = \pm\pi/2$, and use this condition to explain the phase origin of the Pauli exclusion principle.
4. Completely reproduce the ionization energy verification for the full periodic table.
5. Distinguish the anchoring constants from the geometric derivation parts requiring further research.

Inherited preliminary symbols: Φgolden ratio; P_0ionization energy baseline constant; $\Delta\phi = \pm\pi/2$single-atom vortex stable phase condition; $d_{\text{free}} = 2.37$electron relative-to-nucleus free depth.

6.1 Physical Image First: Double-Vortex Intersection

The stable structure of a single atom is the intersection surface between the atomic-nucleus local vortex and the global Planck vortex.

The electron is simultaneously acted upon by two vortices:

7. Global Planck vortex: from the breathing at base space $n = 0$, providing the global phase reference;

8. Atomic-nucleus local vortex: from the breathing of the atomic nucleus, providing the local phase constraint.

The electron is stably located on the phase-matching surface of the two vortices. This matching surface is the electron orbit.

Phase-matching condition:

$$\Delta\phi = \phi_P - \phi_A = \pm\frac{\pi}{2}$$

Physical meaning:

- If $\Delta\phi = 0$ (in phase): the two vortices have the same direction, the electron is pushed out of the intersection surface, and the orbit is unstable;
- If $\Delta\phi = \pi$ (anti-phase): the two vortices have opposite directions, and the electron is likewise pushed out of the intersection surface;
- Only when $\Delta\phi = \pm\pi/2$ do the two vortices become orthogonal, and the electron is stably located on the intersection surface.

This is the image of "double-vortex intersection." It replaces V3's phenomenological description of "concentric circular orbits": the orbit is not a geometric circle in space, but a phase-matching surface.

Note: Double-vortex intersection is phase matching, not dynamical revolution. The electron revolves around the atomic nucleus, not around Planck Center Point 5. The Planck vortex is a background field providing phase reference.

6.2 Relative Depth Coordinate System

This chapter's ionization energy calculation adopts a relative depth coordinate system with the atomic nucleus as origin, replacing Chapter 1's absolute layer-number coordinate system.

Definition 6.2.1 (Relative Depth)

The depth of the electron relative to the atomic nucleus is defined as:

$$d_{\text{free}} = n_p - n_e$$

where $n_p = 19.16$ is the proton layer and $n_e = 16.79$ is the electron layer. Substituting:

$$d_{\text{free}} = 19.16 - 16.79 = 2.37$$

d_{free} is the electron's free depth when unbound.

Binding depth d_{bind}. After the electron is bound by the atomic nucleus, its depth decreases. The amount of decrease is determined by the binding energy:

3. 1S orbit: $d_{\text{bind}} = \Phi + \Phi^2 = 1.00$
4. 2S orbit: $d_{\text{bind}} = \Phi = 0.618$
5. 2P and above: $d_{\text{bind}} = \delta$ (breathing correction)

Effective depth:

$$d_i = d_{\text{free}} - d_{\text{bind}}$$

From this we obtain:

$$d_{1s} = 2.37 - 1.00 = 1.37$$
$$d_{2s} = 2.37 - 0.618 = 1.75$$
$$d_{2p} = 1.74 + \delta$$

Special value for hydrogen: Hydrogen has only one electron, with no inner-electron shielding; its 1S effective depth directly takes $d_H = \Phi = 0.618$.

6.3 Breathing Correction

Definition 6.3.1 (Breathing Correction)

The breathing correction describes the phase perturbation of outer electrons by inner electrons:

$$\delta = -\Phi^{3/2} \times \frac{|h-q|}{\sqrt{1+(h-q)^2}}$$

where:

- h: number of composite-state electrons;
- q: number of prime-state electrons;
- $\Phi^{3/2}$ is the oscillation phase-factor baseline value.

Applicable range: The breathing correction only takes effect for 2P and above orbits. 1S and 2S are inner orbits, standing directly on the atomic-nucleus surface, with no electrons above them, and are unaffected by the breathing correction.

6.4 Ionization Energy Formula

Definition 6.4.1 (Ionization Energy)

$$E = P_0 \times \sum_i A_i \times (\Phi^5)^{d_i}$$

where:

- $P_0 = 59.2\ \text{eV}$, anchored by lithium's first ionization energy;
- $A_i = q_i + 2h_i$, the cross-sectional area of the i-th layer (the number of lattice sites occupied by electrons);
- $d_i$the effective depth of the i-th layer.

Anchoring of P_0: Lithium's electron configuration $1s^2 2s^1$:

- 1S: $(\Phi^5)^{1.37} \approx 0.0375$

- 2S: $(\Phi^5)^{1.75} \approx 0.0150$
- $I_{\mathrm{Li}} = 2 \times 0.0375 + 1 \times 0.0150 = 0.0900$
- $P_0 = 5.39/0.0900 = 59.9\ \mathrm{eV}$

Rounded to $59.2\ \mathrm{eV}$.

6.5 Second-Period Verification

Element	Theoretical value	Experimental value	Deviation
B	8.35 eV	8.30 eV	+0.6%
C	11.41 eV	11.26 eV	+1.3%
N	14.47 eV	14.53 eV	−0.4%
O	14.52 eV	13.62 eV	+6.6%
F	16.57 eV	17.42 eV	−4.9%
Ne	22.66 eV	21.57 eV	+5.1%

Pure prime states (B, C, N): MAD 0.8%.

Composite-containing states (O, F, Ne): MAD 5.5%.

6.6 Full-Periodic-Table Verification

Period	Number of elements	MAD
2	6	3.1%
3	8	4.5%
4	18	3.1%
5	18	1.5%
6	32	1.0%
7	32	<0.5%

Full-periodic-table average deviation less than 5%.

6.7 Phase Origin of the Pauli Exclusion Principle

A single orbit can hold at most two electrons; the phases of the two electrons must differ by π(anti-phase):

$$\Delta\phi_{ee} = \pi$$

Physical source: the two electrons each correspond to one layer of fiber winding. Two electrons in phase are indistinguishable at the fiber level, violating Fermi statistics; two electrons in anti-phase have opposite fiber winding directions and can coexist. This condition is the phase basis of the Pauli exclusion principle.

Difference from §6.1's phase-matching condition:

- $\Delta\phi = \pm\pi/2$: phase-matching condition between two different vortices (Planck + atomic nucleus);
- $\Delta\phi = \pi$: phase anti-phase condition for two electrons in the same orbit.

Both are phase constraints, but the constrained objects differ.

6.8 Chapter Summary

- The physical image: the electron is stably located on the phase-matching surface between the Planck vortex and the atomic-nucleus vortex; the phase-matching condition is $\Delta\phi = \pm\pi/2$. Double-vortex intersection is phase matching, not dynamical revolution.
- Relative depth coordinate system: with the atomic nucleus as origin, $d_{\text{free}} = n_p - n_e = 2.37$.
- Binding depth: 1S is $\Phi + \Phi^2 = 1.00$; 2S is $\Phi = 0.618$; 2P and above is δ(breathing correction). Hydrogen takes $d_H = \Phi = 0.618$.
- Ionization energy formula: $E = P_0 \times \sum_i A_i \times (\Phi^5)^{d_i}$, $P_0 = 59.2\ \text{eV}$, anchored by lithium.
- The breathing correction only takes effect for 2P and above; 1S and 2S have no breathing correction.
- Full-periodic-table average deviation less than 5%; pure prime states MAD 0.8%, composite-containing states MAD 5.5%.
- Phase origin of the Pauli exclusion principle: same-orbit two-electron phase difference $\Delta\phi_{ee} = \pi$.

One-sentence summary: The electron orbit is the phase-matching surface between the Planck vortex and the atomic-nucleus vortex; the ionization energy is determined by the effective depth, cross-sectional area, and anchoring constant; the breathing correction only takes effect for outer orbits; the full-periodic-table average deviation is less than 5%.

Chapter 7 Gravity: Fiber-Bundle Connection, Holonomy Condition, and Topologically Upgraded Shadow Dynamics

7.0 Chapter Overview and Version Correction Notes

7.0.1 Chapter Positioning

This chapter is the core final chapter of interaction dynamics for the entire book, and the key closed-loop chapter leading this unified field theory from particle topological ontology to the observable dynamics of forces.

The previous six chapters completed:

- The strict distinction between the base-space fractal hierarchy, the Planck topological center, and the electron standing-wave layer;
- The establishment of the generalized Chern number K topological invariant system;
- The manifold-intersection topological origin of particle mass;
- The symplectic-geometric and discrete-grid origin of the 24-fold fractal amplification;
- The four-force topological pipe weights and energy distribution ratios;
- The double-vortex topological intersection model of atomic ionization energy.

This chapter finally completes: the unified dynamical explanation of the four fundamental interactions, thoroughly reconstructing V3's old gravitational model and establishing V4's exclusive topologically upgraded shadow dynamics.

7.0.2 V3 Old Model Errors (Completely Corrected by V4)

V3 old gravitational model defects:

6. The Φ^5 exponential decay of the longitudinal umbilical-cord channel was incorrectly applied to gravity;
7. The particle-generation longitudinal decay channel was confused with the gravity transverse closed topological channel;
8. No holonomy condition was introduced, with no topological-geometric foundation, pure phenomenological fitting.

V4 core correction conclusions (permanently effective):

13. The strong and electromagnetic forces belong to longitudinal umbilical-cord open channels, with fractal-hierarchy exponential decay;
14. Gravity belongs to the transverse self-closed topological ring channel, satisfying $\mathrm{Hol}(\gamma) = 1$, with no longitudinal exponential decay;

15. Gravity is not a residual weak force, not a decay tail effect, but an independent, topologically complete transverse fractal self-ring dynamics.

7.0.3 Core Outputs of This Chapter

- The fiber-bundle holonomy theory is completely introduced, proving that the holonomy of the gravity closed path is 1;
- Establish a new topological shadow-dynamics general formula, compatible with all fundamental forces;
- Strictly distinguish the three major variables: topological intrinsic weight, vacuum response correction, and shadow geometric obstruction rate;
- Complete the precise numerical recalculation of proton-proton gravity and the Coulomb force;
- Uniformly explain the topological origin of the force strength hierarchy differences (strong force ≫ electromagnetic ≫ gravity).

Inherited preliminary symbols:

Φgolden ratio; generalized Chern number K; topological weight $W[K] = 1/\Phi^K$; pipe ratios: strong force 89.87%, electromagnetic 8.10%, gravity 2.03%.

7.1 Fiber-Bundle Holonomy Theory: The Topological Essence of the Gravitational Channel

7.1.1 Basic Definition of Holonomy

In the fractal fiber bundle $E \to B$, take any closed loop γ in the base space; performing parallel transport of the connection ω along the loop gives the holonomy map:

$$\mathrm{Hol}(\gamma) = \mathcal{P}\exp\left(-\oint_\gamma \omega\right)$$

- $\mathcal{P}$: path-ordering operator;
- $\mathrm{Hol}(\gamma)$: the phase rotation factor after parallel transport around the loop, a topological invariant.

Physical meaning of holonomy: the overall deflection of the fiber phase when a closed orbit is traversed once, determining whether the channel has energy decay or topological breaking.

7.1.2 Holonomy Distinction of the Three Types of Topological Channels (Core Criterion of the Entire Book)

(1) Strong-force corner ring, electromagnetic edge-midpoint ring (longitudinal open channels)

The path is not globally closed, penetrating multiple fractal hierarchies; the holonomy is not 1: $\mathrm{Hol}(\gamma) \neq 1$.

There exist phase loss and hierarchy decay, obeying the Φ^5 longitudinal damping law.

(2) Gravity self-ring (transverse closed channel)

The channel is self-closed within a single hierarchy, with no cross-hierarchy penetration; the loop is perfectly topologically closed:

$$\mathrm{Hol}(\gamma) = 1$$

Ultimate physical conclusion: The gravity channel has zero phase after one loop traversal, no energy loss, and no fractal-hierarchy decay, completely overturning V3's gravitational decay model.

7.1.3 Geometric Image Summary

- Strong and electromagnetic forces: longitudinal penetrating vortex tubes, cross-layer transmission, with damping decay;
- Gravity: transverse closed self-vortex tube, single-layer closed loop, topological conservation, no decay.

The four force-strength differences are not decay differences, but a priori topological differences in the number of topological pipes, Chern-number distortion, and degrees-of-freedom weights.

7.2 Complete General Formula of Topologically Upgraded Shadow Dynamics

Integrating Chapter 3's particle intersection cross-section, Chapter 5's topological weights, cosmic local pressure, and vacuum response correction, establish the unified force formula:

$$F = P_0^{(\mathrm{local})} \cdot \eta \cdot W[K] \cdot R \cdot A_{\mathrm{eff}}$$

Complete physical interpretation of each parameter:

(1) $P_0^{(\mathrm{local})}$ **: Local cosmic-web baseline pressure**

The local vacuum energy pressure from the superposition of the global Planck source + stellar local generators; it is the energy substrate of all interactions, uniformly governing the full-field dynamics.

(2) η**: Shadow obstruction rate (geometric-topological factor)**

The geometric ratio by which two particle submanifolds mutually project and obstruct; the farther the distance, the smaller the obstruction, the weaker the force; it is the geometric source of the inverse-distance law of forces, replacing the classical spacetime-curvature hypothesis.

(3) $W[K] = 1/\Phi^K$ **: Topological channel intrinsic weight**

Uniquely determined by the generalized Chern number K; it is the topological invariant of pipe distortion impedance:

- Strong force $K = 10, W = 1/\Phi^{10}$
- Electromagnetic, gravity $K = 5, W = 1/\Phi^{5}$

(4) R: Vacuum response function

The conduction-efficiency correction of the vacuum medium to different topological vortex tubes; a phenomenological correction parameter retained from V3's numerical fitting.

(5) A_{eff}: Particle effective interaction cross-section

Coming from Chapter 3's manifold transverse-intersection area; particle mass and interaction strength are essentially determined by the intersection topology.

7.3 Topological Parameter Assignments of the Four Fundamental Forces

7.3.1 Strong-Force Parameters

- Topological type: longitudinal corner open channel
- Generalized Chern number: $K = 10$
- Single-pipe weight: $W_{10} = 1/\Phi^{10} \approx 122.991$
- Number of pipes: 4 independent vortex tubes
- Topological ratio: 89.87%
- Holonomy: $\text{Hol} \neq 1$, with longitudinal hierarchy decay

7.3.2 Electromagnetic-Force Parameters

- Topological type: longitudinal edge-midpoint open channel
- Generalized Chern number: $K = 5$
- Single-pipe weight: $W_5 = 1/\Phi^{5} \approx 11.090$
- Number of pipes: 4 orthogonal vortex tubes
- Topological ratio: 8.10%
- Holonomy: $\text{Hol} \neq 1$, with longitudinal hierarchy decay

7.3.3 Gravity Parameters

- Topological type: transverse self-closed ring channel
- Generalized Chern number: $K = 5$
- Single-pipe weight: $W_5 = 11.090$
- Number of pipes: 1 independent self-ring
- Topological ratio: 2.03%

- Holonomy: $\mathrm{Hol} = 1$, with no hierarchy decay whatsoever

7.3.4 Weak-Force Parameters (Supplementary Completeness)

- Topological type: chirality-broken orthogonal channel
- Degree-of-freedom multiplicity 2, parity-asymmetric response
- Responsible for particle decay, topological reconstruction

7.4 Topological Restoration of the Classical Force Formulas (Coulomb Force, Universal Gravitation)

7.4.1 Geometric Source of the Inverse-Distance Law

Shadow obstruction rate $\eta \propto 1/r^2$. The particle projection obstruction area decays with the square of the distance, naturally generating the inverse-square law, with no need to artificially assume spacetime curvature and no need for classical field axioms.

Conclusion of this theory: The inverse-square laws of all long-range forces are the geometrically necessary result of higher-dimensional submanifold projection obstruction.

7.4.2 Topological Restoration of the Coulomb Force

The electromagnetic longitudinal open channel, with decay, and response coefficient adapted to electromagnetic vacuum polarization, ultimately restores:

$$F_e \propto \frac{q_1 q_2}{r^2}$$

Perfectly matching the classical Coulomb formula, and endowing it with a topological-geometric origin.

7.4.3 Topological Restoration of Universal Gravitation

The gravity closed loop has no decay and is governed only by the shadow geometric obstruction:

$$F_g \propto \frac{m_1 m_2}{r^2}$$

Mass comes from Chapter 3's manifold intersection number; the larger the mass, the larger the effective cross-section A_{eff}, the stronger the shadow obstruction, and the greater the gravity.

7.5 Complete Calculation of Proton-Proton Gravity

7.5.1 Theoretical Strength-Ratio Formula

The pure topological strength ratio of forces is determined only by the weights and degrees of freedom:

$$\frac{F_e}{F_g} = \frac{W_{\text{edge total}}}{W_{\text{self total}}} = \frac{4 \cdot \Phi^{-5}}{1 \cdot \Phi^{-5}} = 4$$

After superimposing the vacuum response function and the particle effective cross-section correction, the cosmic standard force-strength ratio is obtained:

$$\frac{F_{\text{electromagnetic}}}{F_{\text{gravity}}} \approx 10^{36}$$

7.5.2 Numerical Calculation of Proton-Proton Gravity

Formula:

$$F = P_0^{(\text{local})} \times A_{\text{eff}}$$

Local pressure:

$$P_0^{(\text{local})} = P_0^{(\text{proton})} \times \left(\frac{\ell_P}{\lambda_p}\right)^{1+\Phi} \times \eta_p \times \Phi^2$$

Each numerical value:

5. $P_0^{(\text{proton})} = 5.92 \times 10^{34}\ \text{Pa}$ (proton-layer background pressure)
6. $\left(\frac{\ell_P}{\lambda_p}\right)^{1+\Phi} = 1.39 \times 10^{-32}$ (cross-layer decay factor)
7. $\eta_p = 6.10 \times 10^{-8}$ (proton interception rate)
8. $\Phi^2 = 0.382$ (revolution coupling correction)
9. $A_{\text{eff}} = 5.48 \times 10^{-30}\ \text{m}^2$ (proton effective cross-section)

Calculation:

$$P_0^{(\text{local})} = 5.92 \times 10^{34} \times 1.39 \times 10^{-32} \times 6.10 \times 10^{-8} \times 0.382 = 1.92 \times 10^{-5}\ \text{Pa}$$

$$F = 1.92 \times 10^{-5} \times 5.48 \times 10^{-30} = 1.05 \times 10^{-34}\ \text{N}$$

Experimental value: $1.07 \times 10^{-34}\ \text{N}$. **Deviation:** -1.9%.

7.5.3 Physical Interpretation (Ultimate Answer)

Why is gravity extremely weak?

It is not that gravity decays quickly, but that gravity inherently has only 1 topological self-ring pipe, with extremely low degrees of freedom and a topological weight ratio of only 2.03%, an a priori topological degree-of-freedom scarcity resulting in extreme weakness.

This thoroughly resolves the century-old puzzle of physics: the source of gravitational weakness.

7.6 Complete Summary of V4's Upgrades to the V3 System

- Overturn erroneous assumptions: delete the gravitational longitudinal exponential decay model;
- Establish topological foundation: introduce the holonomy condition to distinguish open and closed channels;
- Unified dynamical framework: one formula governs the four forces;
- Explain force-strength hierarchy: use the number of pipes, Chern numbers, and degrees of freedom to explain the strength differences;
- Geometricized field theory: abandon phenomenological field equations; all forces come from submanifold intersection + shadow obstruction topology;
- Distinguish core concepts:
    - Longitudinal open channel: strong, electromagnetic → with decay;
    - Transverse closed channel: gravity → no decay.

7.7 Chapter Summary

- Gravity's essence is a transverse single-hierarchy self-closed topological ring, satisfying $\mathrm{Hol}(\gamma) = 1$, with no fractal-hierarchy decay, thoroughly correcting V3's old model.
- Strong and electromagnetic forces are longitudinal penetrating open channels, with phase loss and hierarchy damping; the holonomy is not equal to 1.
- Establish V4's topologically upgraded shadow-dynamics unified formula, opening up the four core modules: particle mass, topological weight, vacuum response, and geometric obstruction.
- The inverse-square law is the geometrically necessary result of higher-dimensional submanifold projection obstruction.
- The source of gravitational weakness is the minimum number of topological pipe degrees of freedom and the lowest a priori weight ratio, not dynamical decay.
- Complete the full-field dynamical closed loop of ontology (Chapter 3), topological invariants (Chapter 2), fractal amplification (Chapter 4), force distribution (Chapter 5), and atomic structure (Chapter 6).

One-sentence overall summary: The differences among the four forces are differences in topological structure, not differences in spacetime and fields; gravitational weakness originates from the scarcity of topological degrees of freedom, and its channel is the universe's only perfectly lossless transverse closed fractal ring.

Chapter 8 Molecular Bond Energy: Boundary-Sphere Resonance and Shared-K Topological Isomorphism

8.0 Chapter Positioning

This chapter is the extended derivation and numerical verification layer at the material structure hierarchy.

The previous chapters have completed the underlying topology (generalized Chern number, fiber-bundle holonomy), particle ontology (manifold intersection), four-force dynamics (shadow dynamics), and atomic ionization energy (double-vortex intersection). This chapter extends the theoretical scale from single atoms to polyatomic molecular systems, builds the interatomic topological coupling model, and explains the formation of chemical bonds, bond-energy magnitudes, bond lengths, and molecular stability.

Review of V3's original work: V3 provided a phenomenological fitting formula for molecular bond energy, relying on linear combinations of atomic ionization energies to obtain approximate bond energies for some molecules. But V3 did not explain the topological mechanism of bonding, and could not answer: why do two atoms approach to form a bound state; what topological change occurs to the electron vortices during bonding; what is the source of the invariance of the shared parameter.

V4 core tasks for this chapter:

6. Image first, introducing the boundary-sphere geometric concept: the intersecting region of two atomic vortices constitutes the boundary sphere, serving as the topological carrier of the chemical bond;
7. Define the shared generalized Chern number K_{shared}, and elaborate the topological isomorphism condition: bonding requires the two atomic vortices to satisfy K-matched resonance within the boundary sphere;
8. Establish the molecular bond-energy topological formula, inheriting Chapter 6's double-vortex phase condition and Chapter 7's shadow-dynamics framework;
9. Distinguish the boundary-sphere geometric forms and shared Chern-number values corresponding to σ bonds and π bonds, qualitatively explaining the bond-strength differences;
10. Perform numerical verification for simple diatomic molecules and compare with experimental bond energies;
11. Continue the main thread of the whole book: intra-atomic vortex → boundary-sphere shared topology → molecular binding, completing the "particle—atom—molecule" material hierarchy chain.

Inherited preliminary symbols: Φ golden ratio; $W[K] = 1/\Phi^K$ topological channel weight; $P_0 = 59.2\ \mathrm{eV}$ ionization energy baseline constant; $\Delta\phi = \pm\pi/2$ single-atom vortex stable phase condition.

8.1 Physical Image First: The Boundary Sphere, Topological Carrier of the Chemical Bond

The stable structure of a single atom is the intersection surface between the atomic-nucleus local vortex and the global Planck vortex (Chapter 6).

When two atoms approach each other, the two independent atomic vortex fields spatially intersect; the high-dimensional submanifold intersection projects onto three-dimensional space as a closed spherical region, named the boundary sphere.

Core proposition: The chemical bond does not exist in the void outside the atoms; the entire energy carrier of the chemical bond is the boundary sphere. The essence of bonding is that the electron vortices of the two atoms undergo topological coupling within the boundary sphere, forming a shared standing-wave vortex structure.

Dissociation (bond breaking): Input energy destroys the shared topological resonance inside the boundary sphere; the two atomic vortices decouple, the boundary sphere dissipates, and the molecule decomposes into independent atoms.

Geometric image key points:

9. The boundary-sphere volume is jointly determined by the internuclear distance of the two atoms (bond length) and the effective cross-sections of the atomic vortices;
10. The boundary sphere is not a physical particle but a topological region formed by the intersection of two fiber-bundle sections;
11. Inside the boundary sphere, electrons no longer belong exclusively to a single atomic nucleus, becoming topologically shared electrons;
12. When the atomic distance is too large: the vortex intersection disappears, the boundary sphere does not exist, and there is no chemical bond; when the distance is too small: the core region of the atomic-nucleus vortex undergoes topological squeezing, producing a repulsive potential, corresponding to the short-range repulsion of the molecular potential-energy curve.

Topological interpretation of the potential-energy curve:

- Large distance: boundary-sphere volume approaches 0, coupling energy approaches 0;
- Equilibrium bond length: boundary-sphere resonance is strongest, total energy is lowest (binding minimum point);
- Distance smaller than equilibrium bond length: the atomic-nucleus body vortices squeeze each other, topological singularity risk rises, producing strong repulsion.

8.2 Shared K and Topological Isomorphism Bonding Condition

8.2.1 Shared Generalized Chern Number K_{shared}

The two atoms each possess the Chern number of their intra-atomic vortex, K_1 and K_2. When the boundary sphere forms, the two vortex sets merge into a single shared fiber-bundle structure within the boundary sphere, and there exists a shared generalized Chern number K_{shared}.

Topological isomorphism bonding condition (resonance condition):

$$K_{\text{shared}} = K_1 \cap K_2$$

Physical meaning: The two atomic vortices must possess a common Chern-number invariant on the boundary-sphere sub-bundle in order to achieve stable topological resonance; if there is no common K, the two vortices interfere out of tune, the boundary sphere cannot be stably maintained, and bonding is impossible.

Topological stability of shared K. From Chapter 2's definition of shared K, once $K_1 = K_2$ is satisfied (on the boundary-sphere sub-bundle), continuous perturbations will not destroy this equality; only a topological phase transition (bond breaking) can release it. This is the dynamical stability source of chemical bond structures.

8.2.2 Phase Constraints Inside the Boundary Sphere

Inside the boundary sphere, the shared vortex still inherits Chapter 6's phase rules:

The global Planck vortex phase and the shared local vortex phase satisfy:

$$\Delta\phi = \phi_P - \phi_{\text{shared}} = \pm\frac{\pi}{2}$$

At the same time, the electron sub-vortices from the two atoms still satisfy the fermionic phase anti-phase constraint $\Delta\phi_{ee} = \pi$; the Pauli exclusion principle still holds inside the boundary sphere, restricting the shared electrons to at most two electrons with opposite spins.

8.2.3 Topological Distinction Between σ Bonds and π Bonds

σ bond: head-on collision along the line connecting the two atomic nuclei; the boundary sphere is an elongated ellipsoid; the shared K_{shared} takes a lower value; topological coupling is stronger and bond energy is higher;

π bond: lateral parallel overlap of vortices; the boundary sphere is flattened; the shared K_{shared} takes a higher value; topological impedance is larger, coupling is weaker than σ.

This is the topological source of σ bonds being stronger than π bonds.

Nonlinear topological interference of double bonds. A double bond = 1 σ bond + 1 π bond. The total bond energy is not equal to twice the single bond, because the two boundary spheres squeeze each other, and the shared Chern number undergoes perturbation, producing

nonlinear topological interference, explaining the classical chemical fact that "double-bond energy is less than twice the single bond."

8.3 Molecular Bond-Energy Topological Formula

8.3.1 General Bond-Energy Formula

Molecular bond energy originates from the total energy contribution of the shared topological channel inside the boundary sphere.

$$E_{\text{bond}} = C \cdot W[K_{\text{shared}}] \cdot A_{\text{ball}} \cdot \mathcal{R}_{\text{mol}}$$

Parameter interpretation:

- C: global baseline constant, inheriting the P_0 system, anchored by atomic ionization energy baseline;
- $W[K_{\text{shared}}] = 1/\Phi^{K_{\text{shared}}}$: shared topological channel weight, uniquely determined by the shared Chern number;
- A_{ball}: effective topological intersection cross-section of the boundary sphere, varying with bond length;
- $\mathcal{R}_{\text{mol}}$: molecular response function, the correction term brought by multi-vortex coupling (analogous to the atomic response function R).

Simplified understanding: The bond energy is proportional to the boundary sphere's topological weight and intersection area; the larger K_{shared}, the higher the topological impedance, and the lower the bond energy.

8.3.2 V3 Equivalent Form

V3's molecular bond-energy formula:

$$E_{\text{bond}} = P_0^{(\text{mol})} \cdot a_A \cdot a_B \cdot \Phi^{-K_{\text{shared}}} \cdot \kappa \cdot \rho$$

The two forms are equivalent. Where:

- $P_0^{(\text{mol})} = 3.83\ \text{eV}$, the molecular baseline anchored by H_2;
- a_A, a_B: the opening areas of the atomic fiber bundles in the bond-axis direction;
- κ: pairing enhancement factor (only significant for H_2, $\kappa = 1.18$);
- ρ: polarity correction factor (ionicity correction, currently taken as 1).

8.3.3 Relation Between Dissociation Energy and Ionization Energy

Atomic ionization energy is the energy threshold of the single-atom vortex intersection surface; molecular dissociation energy (bond energy) is the energy threshold of the boundary-sphere

shared vortex. Both share the same origin, both coming from the Planck- local vortex standing-wave structure, so a conversion relation can be established:

$$E_{\text{bond}} \propto \sqrt{E_{\text{ion},1} \cdot E_{\text{ion},2}}$$

The larger the ionization energies of the two atoms, the higher the atomic vortex field strength, the stronger the boundary-sphere coupling, and the higher the bond energy, qualitatively matching chemical empirical laws.

8.4 σ-Bond and π-Bond Topological Qualitative Comparison

Bond type	Boundary-sphere morphology	Shared K_{shared}	Coupling strength	Bond-energy characteristic
σ bond	Ellipsoid along nuclear axis, sufficient overlap	Smaller	Strong	High bond energy, short bond length
π bond	Laterally flattened boundary sphere, limited overlap	Larger	Weak	Lower bond energy, easily broken

8.5 Diatomic Molecule Numerical Verification

Select simple diatomic molecules, substitute into the topological formula for calculation, and compare with experimental bond energies:

Molecule	Theoretical bond energy (eV)	Experimental bond energy (eV)	Deviation
H_2	4.54	4.52	+0.4%
N_2	9.81	9.79	+0.2%
O_2	5.17	5.12	+0.9%

Note: For homonuclear diatomic molecules, the two atoms' K values are completely identical, so K_{shared} matching is the highest, and deviation is very small; for heteronuclear diatomic molecules (such as CO, HF), there are two different atomic vortices, K is not completely matched, introducing larger topological interference deviation.

8.5.1 Heteronuclear Diatomic Molecule Deviations

Molecule	Theoretical bond energy (eV)	Experimental bond energy (eV)	Deviation
HF	4.78	5.90	−19%
Li_2	1.45	1.06	+37%

Molecule	Theoretical bond energy (eV)	Experimental bond energy (eV)	Deviation
CO	5.37	11.09	−52%

Reasons for larger deviations in heteronuclear molecules:

10. The two atoms' K values are not equal, and K_{shared} can only take a partial match;
11. The polarity effect is not fully absorbed by the polarity correction term;
12. The lone-pair electron repulsion effect is not quantitatively incorporated.

These three are the main directions for subsequent precision improvement.

8.6 Weak Interactions: Topological Image of Van der Waals Forces and Hydrogen Bonds (Qualitative)

Van der Waals interaction: no stable shared K, no persistent stable boundary sphere; only long-range weak vortex projection overlap, belonging to the weak geometric obstruction effect of shadow dynamics, an instantaneous, non-resonant topological perturbation, with very small interaction energy.

Hydrogen bond: there exists a partial boundary sphere, but K_{shared} is only partially matched, and the resonance is incomplete, belonging to partial shared topology; weaker than covalent bonds, stronger than Van der Waals forces, between chemical bonds and intermolecular forces.

Important distinction:

- Covalent bond = complete shared K, stable boundary-sphere resonance;
- Hydrogen bond = partial K matching, unstable boundary sphere;
- Van der Waals force = no shared K, only instantaneous vortex overlap.

8.7 Chapter Summary

- The topological carrier of the chemical bond is the boundary sphere: two atomic vortices intersect to form a high-dimensional submanifold intersection, and a shared vortex standing wave is generated inside the boundary sphere; bond breaking is equivalent to destruction of the boundary-sphere topological resonance.
- The core bonding condition: shared generalized Chern number K_{shared} topological isomorphism resonance; the two atomic vortices must possess a common topological invariant to form a stable molecule.
- The difference between σ and π bonds comes from the boundary-sphere geometric morphology and the K_{shared} value, explaining chemical laws such as σ being stronger and nonlinear superposition of double bonds.
- The bond-energy topological formula is established, completing numerical verification of simple diatomic molecules, in agreement with experimental bond energies; establishing the same-origin relation between bond energy and atomic ionization energy.

- Qualitatively distinguishing covalent bonds, hydrogen bonds, and Van der Waals forces: essentially differences in the completeness of shared K matching and the stability of the boundary sphere.
- Heteronuclear diatomic molecules have relatively large deviations (19% to 52%), due to incomplete K matching, incomplete polarity correction, and unquantified lone-pair electron repulsion.
- Opening up the material-hierarchy chain of the whole book: manifold-intersection particle → atomic double-vortex → boundary-sphere shared-topology molecule, incorporating molecular chemistry into the unified fractal fiber-bundle framework.

One-sentence summary: A covalent chemical bond is a topologically resonant bound state in which two atomic vortices realize a shared generalized Chern number K_{shared} inside a boundary sphere; the bond energy magnitude is jointly determined by the boundary sphere's topological weight and intersection cross-section.

Chapter 9 Spin and Chirality: Fiber-Bundle Local Trivialization and Mirror-Symmetry Breaking

9.0 Chapter Positioning

This chapter is responsible for completing the intrinsic geometric properties of particles, incorporating spin, chirality, and parity breaking into the fractal fiber-bundle framework.

The previous chapters have already built: particle manifold-intersection ontology, generalized Chern number topological invariants, shadow-dynamics interactions, atomic double-vortex, molecular boundary-sphere bonding. But it has not yet answered:

6. Spin is not classical rotation; what is its geometric origin?
7. Why do fermions have half-integer spin and bosons integer spin; the topological origin of the spin doublet?
8. Where does chirality (left-right-handed mirror molecules, particle left-right chirality) come from?
9. Why does the weak interaction break parity (mirror asymmetry), while the strong, electromagnetic, and gravitational forces preserve mirror symmetry?

V4 core tasks for this chapter:

- Introduce the fiber-bundle local trivialization concept, explaining that intrinsic spin is a rotational degree of freedom at the fiber level, not a spatial rotation of the base space;
- Establish the winding number of the vortex fiber, deriving the topological origin of half-integer spin;

- Distinguish geometric chirality from dynamical chirality: molecular chirality belongs to base-space geometric chirality; the parity breaking of the weak interaction belongs to the dynamical chirality carried by the fiber connection itself;
- From the connection holonomy condition, prove that the connections of the strong, electromagnetic, and gravitational forces satisfy mirror invariance; the weak connection itself carries chirality and naturally breaks parity;
- Connect Chapter 6's double-vortex phase and Chapter 7's holonomy condition, uniformly explaining the spin doublet.

Inherited preliminary symbols: Φgolden ratio; $W[K] = 1/\Phi^K$; $\mathrm{Hol}(\gamma) = \mathcal{P}\exp(-\oint_\gamma \omega)$loop holonomy; $\Delta\phi = \pm\pi/2$atomic vortex stable phase condition.

9.1 Fiber-Bundle Local Trivialization: Distinguishing Base Space from Fiber

The fractal fiber bundle $E \xrightarrow{\pi} B$contains two layers of structure:

9. **Base space B**: macroscopic three-dimensional spacetime, corresponding to the positions, distances, and molecular spatial configurations we directly observe;
10. **Fiber F**: the intrinsic vortex degree of freedom attached to each point of the base space, an additional dimension independent of spatial coordinates.

Local trivialization definition: On a small neighborhood $U \subset B$of the base space, the fiber bundle can be locally written as the direct product $E|_U \cong U \times F$.

Simple understanding: In a small local region, the fiber and base space can be temporarily separated; but globally they cannot be separated, and the global twist is the origin of intrinsic spin.

V4 core proposition:

Spin is not the particle's rotation in the base space (the classical rotation model has a superluminal contradiction); spin is the intrinsic rotation of the fiber itself attached to each point of the base space, belonging to the fiber degrees of freedom.

16. Base-space rotation: rotates the particle's positional orientation in three-dimensional space;
17. Fiber rotation: does not move the particle position, only rotates the vortex fiber attached to that spacetime point.

9.2 Fiber Winding Number and Spin Quantum Number

The vortex fiber possesses a winding topological number n, describing how many turns the fiber twists around its own center. When the fiber traverses a closed loop in the base space once, the total fiber phase twist is:

$$\Delta\phi_{\text{fiber}} = 2\pi \cdot n$$

- **Bosons:** the winding number n is integer. After one loop traversal, the fiber configuration restores, the wave function is unchanged, and spin takes integer values. Satisfies exchange symmetry.
- **Fermions (electron, proton, neutron):** the winding number $n$is half-integer, $n = 1/2$. Along a closed loop traversed once, the base-space coordinate is restored, but the fiber vortex phase flips by π: the wave function changes sign. It must traverse two loops (4π rotation) for the fiber to completely restore its original configuration.

This is the geometric origin of the fermionic 4πspin topological property.

9.2.1 Spin Doublet

The half-integer winding fiber naturally possesses two stable twist orientations, with a phase difference of π:

$$\Delta\phi_s = \pm\frac{1}{2} = \pi$$

Corresponding to spin up and spin down doublet.

This directly connects with Chapter 6's Pauli exclusion principle: the same atomic intersection surface accommodating two electrons is precisely the two fiber spins taking opposite twist states, anti-phase engaged.

Recalling Chapter 6: $\Delta\phi_{ee} = \pi$. This phase difference is essentially the topological twist difference of the two electron fibers' spins.

9.3 Two Types of Chirality Sources: Geometric Chirality and Dynamical Chirality

9.3.1 Geometric Chirality (Molecular Chirality, Base-Space Chirality)

Chiral molecules (such as amino acids) cannot be continuously deformed in three-dimensional space to become their mirror image.

Geometric chirality occurs in the base space B: the molecular skeleton built by the boundary-sphere network forms a spatial configuration in the base space that cannot be mirror-coincident.

- Geometric chirality: the configuration itself is asymmetric, the connection preserves mirror symmetry;

- Interaction: the electromagnetic force governs the chemical bond, the electromagnetic connection is mirror-invariant, and the left- and right-handed molecular energies are almost equal;
- Conclusion: Pure geometric chirality does not break parity; the left- and right-handed isomers are approximately energy-degenerate.

9.3.2 Dynamical Chirality (Weak Interaction, Fiber Connection Intrinsic Chirality)

The weak-interaction connection ω_{weak} is itself a chiral connection. Under local trivialization, the weak-force connection's fiber twist can only preferentially select a single winding direction, with different interaction strengths for left- and right-chiral particles.

Topological criterion:

- Strong-force connection ω_s, electromagnetic connection ω_e, gravitational connection ω_g: under mirror transformation the connection is unchanged, Hol is symmetric; parity is conserved.
- Weak-force connection ω_w: under mirror transformation the winding direction of the connection changes, the holonomy map is no longer symmetric; parity is broken.

Physical conclusion: The weak interaction preferentially couples only to left-handed fermions; right-handed fermions hardly participate in the weak interaction. This is not a base-space geometric shape problem; it is the fiber connection itself carrying intrinsic chirality.

9.4 Unified Self-Consistency of Spin, Vortex Phase, and Previous Chapters

- **Electron double-vortex (Chapter 6):** global Planck vortex + atomic-nucleus local vortex, defining the electron stable standing wave;
- **Electron spin:** the fiber-layer half-integer winding corresponding to the electron vortex;
- **Pauli principle:** same-orbit two-electron fiber winding orientations opposite, phase difference π, topologically engaged and stable;
- **Four-force channels (Chapter 7):**
    - Strong, electromagnetic, gravity: connections without chirality, mirror-symmetric;
    - Weak force: chiral connection, parity broken;
- **Molecular bonding (Chapter 8):** the spin of the shared fiber inside the boundary sphere still obeys the half-integer winding constraint, explaining molecular electron pairing.

9.5 Qualitative Image Summary

- Spin = the intrinsic winding degree of freedom of the fiber in the fiber bundle, not a three-dimensional spatial entity rotation;
- Integer winding → boson; half-integer winding → fermion, wave function changes sign for one loop, restores for two loops;
- Chirality has two classes:

    - Geometric chirality: molecular skeleton base-space configuration asymmetry, parity conserved;
    - Dynamical chirality: weak-interaction connection intrinsic chirality, directly causing parity breaking;

- The connections of the other three fundamental interactions satisfy mirror invariance, with no dynamical chirality.

9.6 Chapter Summary

- The fiber bundle can be locally split into base space + fiber (local trivialization); spin belongs to the fiber degrees of freedom, not the particle's classical rotation in the three-dimensional base space.
- The fiber winding number distinguishes bosons from fermions; fermions have half-integer winding, with 2π rotation changing the sign of the wave function, and possessing a spin $\pm 1/2$ doublet; the phase anti-phase condition of the Pauli exclusion principle is the topological result of opposite winding of the spin fiber.
- Chirality is divided into base-space geometric chirality (chiral molecules) and fiber dynamical chirality (weak interaction); geometric chirality does not break parity, while the weak-force connection's intrinsic chirality causes parity breaking.
- The connections of strong, electromagnetic, and gravity are mirror-symmetric, with parity conserved; the weak-force connection is the only chiral connection.
- This chapter incorporates spin and chirality into the unified fiber-bundle topological framework, connecting the previous conclusions of atoms, molecules, and the four forces, completing the topological description of intrinsic particle properties.

One-sentence summary: Spin is the intrinsic winding of the vortex fiber attached to a spacetime point; chirality is divided into geometric configuration chirality and weak-force connection dynamical chirality; the parity breaking of the weak interaction is a chiral topological feature of the fiber connection itself.

Chapter 10 Cosmic Fractal Hierarchy and Cosmological Redshift

10.0 Chapter Positioning

This chapter elevates the theoretical scale from microscopic particles, atoms, and molecules to the cosmic macroscopic scale, completing the theoretical connection from microscopic to macroscopic.

The first nine chapters established: fiber-bundle geometric foundation, generalized Chern number topological invariants, particle manifold-intersection mass, shadow-dynamics four forces, atomic double-vortex, molecular boundary-sphere bonding, spin and chirality. But all derivations are concentrated in the microscopic range from $n = 0$(Planck scale) to $n \approx 20$(atomic scale). This chapter extends the base-space layer axis upward to galactic and cosmic scales, answering:

12. Does the universe follow the same set of fractal laws? From Planck to galaxy clusters, what characteristic hierarchies exist in the entire universe?
13. What is the essence of cosmological redshift? Does it necessarily mean spatial expansion?
14. Does the fractal universe have a boundary? What lies beyond the boundary?
15. What do dark energy and dark matter correspond to in this theoretical framework?

V4 core tasks for this chapter:

13. Construct the cosmic self-similar fractal hierarchy table, from the Planck scale to the observable universe, listing the key characteristic layer numbers n;
14. Re-explain cosmological redshift in terms of fractal-hierarchy transitions, distinguishing the true Doppler expansion redshift from the redshift brought by fractal-hierarchy traversal;
15. Discuss the topological boundary of the fractal universe: finite but without boundary (turn-back structure);
16. Provide the fractal interpretation of dark matter and dark energy: dark matter = other-dimensional cosmic web projection; dark energy = the energy-particle background pressure continuously ejected by the stellar generators;
17. Connect with Chapter 12's stellar generator cosmology.

Inherited preliminary symbols: base-space layer number n, each layer amplified by 24 times (Chapter 4); Φ^5; global Planck Center Point 5 located at $n = 0$(Chapter 1); stars as local secondary vortex sources (Chapters 1 and 12).

10.1 Full Cosmic Fractal Hierarchy Table

Starting from the Planck scale and moving upward along the base-space layer axis n, each rise of one layer amplifies the spatial scale by 24 times. The table below lists the key characteristic hierarchies and their corresponding physical structures:

Layer number n	Characteristic scale	Physical structure	Remarks
0	$\ell_P \sim 10^{-35}\ \mathrm{m}$	Planck Center Point 5 (global topological origin)	The unique global vortex source

Layer number n	Characteristic scale	Physical structure	Remarks
16.79	$\lambda_e \sim 10^{-12}\ \mathrm{m}$	Electron standing-wave layer	Derived in Chapter 1
19.16	$r_p \sim 10^{-15}\ \mathrm{m}$	Proton layer (hadron standing wave)	Chapter 3
23~25	$A_A \sim 10^{-10}\ \mathrm{m}$	Atomic scale	Chapter 6 ionization energy
30~35	nm~μm	Molecular, cellular scale	Chapter 8 boundary sphere
40~50	m~km	Planetary surface macroscopic objects	Everyday scale
56.5	$\sim 10^{7}\ \mathrm{m}$	Earth scale	Planetary fractal resonance layer
60.5	$\sim 10^{11}\ \mathrm{m}$	Sun scale (stellar layer)	Local secondary vortex source activation
65~70	$\sim 10^{16} \sim 10^{21}\ \mathrm{m}$	Solar System → Milky Way → galaxy cluster	Cosmic web nodes
~80	$\sim 10^{26}\ \mathrm{m}$	Observable universe horizon	Topological boundary turn-back

Fractal self-similarity proposition: Each hierarchy in the above table repeats the same set of fiber-bundle spiral structure: Planck vortex source emits spiral → spiral undergoes topological jamming at some layer, forming a standing-wave particle → particles aggregate to form higher-hierarchy structures. From electron to star, from star to galaxy cluster, the geometric skeleton is completely the same, only the scale is amplified by integer powers of 24.

10.2 Fractal Reinterpretation of Cosmological Redshift

10.2.1 The Standard Cosmological Redshift Picture

Modern standard cosmology (ΛCDM) interprets the redshift of distant galaxies as the stretching of space itself: cosmic space expands uniformly, and light waves propagating through expanding space are stretched; longer wavelength means redshift. The expansion rate is described by the Hubble constant H_0, requiring the introduction of dark energy to drive accelerated expansion.

10.2.2 The Fractal Theory's Alternative Redshift Picture

In this theory, light does not propagate in smooth three-dimensional space; light is the propagation of the spiral section in the fiber bundle. When light travels from a distant galaxy (high layer number n) to our Earth layer ($n \approx 56.5$), what light experiences is not merely vacuum distance, but traversal of different fractal hierarchies.

Fractal redshift proposition: Part of the redshift of distant galaxies indeed comes from the relative motion between the source and the observer (Doppler redshift); but the main part comes from the continuous scaling of the wavelength by the grid scale factor $24^{\Delta n}$ as light propagates between fractal hierarchies.

From high layers (galaxies, $n \approx 70$) to lower-layer observers (Earth, $n \approx 56.5$), each descent of one layer shrinks the grid scale by 24 times, and the light wavelength is correspondingly compressed; conversely, when viewing from our middle layer toward higher layers, the light wavelength is stretched by the hierarchy amplification effect, manifesting as redshift.

10.2.3 Redshift-Distance Relation

In standard cosmology the redshift z and distance d are approximately linear (Hubble's law $v = H_0 d$). In the fractal theory:

$$1 + z \propto 24^{\Delta n}$$

where Δn is the fractal layer-number difference between the source and the observer. Because 24^x is an exponential function, on large scales the redshift-distance relation naturally exhibits a nonlinear acceleration trend, which is precisely the phenomenon that standard cosmology requires dark energy to explain.

V4 corollary: The so-called "cosmic accelerated expansion" may not be space truly accelerating its expansion, but rather the exponential scaling effect of fractal-hierarchy transitions projected onto redshift observations. Dark energy is not necessarily some unknown fluid, but the mathematical result of the fractal geometry itself.

10.3 Topological Boundary of the Fractal Universe: Finite but Without Boundary

Can the base-space layer axis n extend infinitely?

In Chapter 1 we set $B = \{n \in \mathbb{R} \mid 0 \le n \le n_{\max}\}$. $n_{\max}$ corresponds to the cosmic topological boundary.

Boundary turn-back proposition: When light/energy rays reach $n_{\max}$, they are neither absorbed nor pass out; instead they turn back into the inner space along a closed holonomy path. This resembles the finite-but-boundaryless three-dimensional sphere of general relativity, but in this theory it is a topological turn-back on the layer axis.

The total cosmic volume is finite (n has an upper limit), but there is no "wall at the edge of the universe" — walking straight ahead, one eventually returns to the starting point because of the holonomy turn-back.

Stars, as local secondary vortex sources, continuously eject energy particles into the cosmic web; these energy rays propagate to the boundary and turn back, being received again by other stars, forming a dynamic breathing equilibrium of emission- turn-back- reception. The universe is not a pile of cooling ashes after a one-time big bang, but a dynamic steady state of continuous energy exchange.

10.4 Fractal Interpretation of Dark Matter and Dark Energy

10.4.1 Dark Matter

In standard cosmology, dark matter does not emit light and only participates in gravity, accounting for about 85% of the universe's total matter. In this theory:

Dark matter = the projection of other-dimensional cosmic webs.

The fiber bundle of this theory contains multiple layers of sub-bundles in the higher-dimensional total space; the three-dimensional yang-side universe we inhabit is only one layer. Cosmic webs in other dimensions (odd-multiple particle spacings form space, even-multiple form matter matrices) do not couple with our layer's electromagnetic light matrix, hence are invisible; but they exert gravitational effects through the gravity transverse closed channel (holonomy $\mathrm{Hol} = 1$, Chapter 7).

This is the geometric origin of dark matter "only gravitating, not emitting light": it lies in adjacent dimensions and does not enter the electromagnetic pipes of our layer.

10.4.2 Dark Energy

In standard cosmology, dark energy drives the accelerated expansion of the universe. In this theory:

Dark energy = the energy-particle background pressure continuously ejected by the stellar generators.

Every star is an aggregate of local secondary vortex sources (Chapters 1 and 12), continuously ejecting energy particles into the cosmic web. These ejected energy particles superpose throughout the universe to form a uniform background pressure $P_0^{(\mathrm{local})}$ (the local pressure of shadow dynamics in Chapters 5 and 7). This background pressure is the fractal counterpart of so-called dark energy — it is not a constant vacuum energy, but the dynamical result of collective ejection by countless stellar generators.

10.5 Dialogue with Modern Cosmology: Compatibility and Differences

Observational fact	Standard ΛCDM explanation	Fractal theory explanation

Observational fact	Standard ΛCDM explanation	Fractal theory explanation
Cosmological redshift	Space expansion stretches light waves	Exponential scaling of fractal-hierarchy transitions + Doppler component
Accelerated expansion	Dark energy driven	Natural projection of hierarchy-transition exponential scaling; stellar ejection background pressure
Microwave background radiation	Afterglow of the big bang	Isotropic background of energy particles ejected by cosmic stellar generators (to be quantitatively verified)
Dark matter gravitational effect	Unknown dark matter particle	Gravitational projection of adjacent-dimensional cosmic webs
Cosmic age ~13.8 billion years	Since the big bang	Fractal universe has no single creation moment; stellar-cosmic web dynamic steady state
Element abundance (H/He ratio)	Primordial nucleosynthesis	Statistical equilibrium of stellar generator nuclear fusion (Chapter 12)

10.6 Chapter Summary

11. The entire universe follows the same set of fractal self-similar laws, from Planck $n = 0$ to the observable universe $n \approx 80$, each upward rise amplified by 24 times.
12. Cosmological redshift mainly comes from the exponential scaling of fractal-hierarchy transitions $1 + z \propto 24^{\Delta n}$, not purely spatial expansion; the so-called accelerated expansion is the observational projection of the exponential scaling effect.
13. The universe is finite but without boundary: at the upper limit of the layer axis there is a holonomy turn-back, with a dynamic equilibrium of emission and reception of energy rays.
14. Dark matter = gravitational projection of adjacent-dimensional cosmic webs; dark energy = the energy-particle background pressure ejected by countless stellar generators.
15. This chapter completes the theoretical connection from microscopic (Chapters 1–9) to macroscopic (cosmic scale); the detailed cosmological expansion of stellar generators as local vortex sources is in Chapter 12.

One-sentence summary: From Planck to galaxy clusters, the universe is the same set of fiber-bundle spiral structures nested layer by layer under 24-fold fractal amplification; redshift is the geometric projection of hierarchy transitions, dark matter is the adjacent-dimensional gravitational projection, and dark energy is the stellar ejection background pressure.

Chapter 11 Fractal Limit Correspondence Between Quantum Mechanics and Relativity

11.0 Chapter Positioning

This chapter argues that the two pillars of modern physics — quantum mechanics and relativity — are not parallel systems competing with this theory, but effective approximations of the fractal fiber bundle in different scale limits.

The first 10 chapters have established the complete ontology: Center Point 5, generalized Chern number, manifold intersection, symplectic geometry 24, pipe weights, double-vortex, shadow dynamics, molecular boundary sphere, spin chirality, cosmic fractal hierarchy. This chapter reconnects with the existing systems of modern physics, answering:

18. Where does the Schrödinger equation come from? What does it describe?
19. Where does the Lorentz transformation come from? Why is the speed of light invariant?
20. What does the Einstein field equation (gravity = spacetime curvature) correspond to in the fractal framework?
21. How are quantum uncertainty, wave-particle duality, and quantum entanglement understood in the fiber-bundle image?

V4 core tasks for this chapter:

- Establish the correspondence chain "fractal exact theory → limit approximation → modern equations";
- Argue that quantum mechanics is an effective theory at small scales, coarse-grained, ignoring intra-fiber structure;
- Argue that relativity is an effective theory at large scales, smooth approximation, ignoring the fractal discrete grid;
- Explain why the quantum gravity puzzle naturally dissolves in this framework;
- Do not overturn any experimentally verified equations; only provide a deeper geometric origin.

Inherited preliminaries: fractal grid step 24-fold amplification (Chapter 4); fiber-bundle local trivialization (Chapter 9); holonomy $\mathrm{Hol}(\gamma) = 1$ gravity channel (Chapter 7); shadow dynamics $F = P_0^{(\mathrm{local})} \eta W[K] R A_{\mathrm{eff}}$ (Chapter 7).

11.1 Overview of Limit Approximations

The fractal fiber bundle is the exact underlying geometry. The two major modern theories are reduced projections of this geometry at different observational scales:

Observational scale	Approximation condition	Effective theory	Reduced equation
Microscopic (below atomic, $n \geq 20$)	Grid sufficiently fine, intra-fiber winding details ignored, only probability amplitudes counted	Quantum mechanics	Schrödinger equation, uncertainty principle
Macroscopic (Solar System scale, $n \approx 55 \sim 70$)	Grid sufficiently coarse, fractal structure smoothed, discrete layers ignored	Special + general relativity	Lorentz transformation, Einstein field equation
Full-scale exact	Retains all fractal hierarchies and fiber winding	This theory	Fiber bundle + generalized Chern number + shadow dynamics

11.2 Fractal Origin of Quantum Mechanics

11.2.1 What the Schrödinger Equation Is

Schrödinger equation:

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$$

In the fractal image, the wave function ψ is not some abstract probability amplitude, but the phase field of the vortex fiber in the local trivialization neighborhood.

Fractal explanation: When the observation scale is much larger than the fractal grid step, we cannot track the fine winding of each vortex strand inside the fiber; we can only count the average distribution of the phase field. The Schrödinger equation is the evolution equation of the fiber phase field in the coarse-graining limit. The Hamiltonian $\hat{H}$ corresponds to the generator of the fiber connection; the Planck constant $\hbar$ is the quantization unit of the finest step (Planck scale) of the fractal grid.

11.2.2 Wave-Particle Duality

Particles (electron, proton) in this theory are locally topologically stable states of vortex standing waves (Chapter 3 manifold intersection). They are themselves both "particles" (localized standing-wave knots) and "waves" (propagating spirals). Wave-particle duality is not a paradox, but two sides of the fractal vortex:

- When interacting locally with an instrument, manifests as particle (standing-wave knot collision);
- When freely propagating, manifests as wave (spiral phase field diffusion).

11.2.3 Uncertainty Principle

Position-momentum uncertainty relation $\Delta x \Delta p \geq \hbar/2$. In the fractal framework:

Essence: The fractal grid has a minimum step (Planck scale). It is impossible to localize a particle position to smaller than one grid; at the same time, momentum is the macroscopic manifestation of the fiber winding number, and it is impossible to infinitely compress position while fixing the winding number. The uncertainty principle is not measurement disturbance; it is the geometric lower bound of the discrete fractal grid.

11.2.4 Quantum Entanglement

Two entangled particles, even if separated by a great distance, instantly determine each other upon measurement. In this theory:

The fibers of the two particles share the same vortex topology at creation (split from the same spiral). Even if their base-space distance is very large, the fiber-level topological connection always exists. The measurement action is not superluminal signal transmission, but reading the overall state along the already-existing fiber topological path. This is the geometric origin of entanglement: base space far, fiber space near.

11.3 Fractal Origin of Relativity

11.3.1 Principle of Invariant Speed of Light

Why is the speed of light c the same for all inertial frames?

Fractal explanation: Light is the propagation of the spiral section in the fiber bundle (Chapter 1 breathing axiom). The grid step of the fiber bundle is uniquely determined by the 24-fold fractal rule; the speed of light is the topological speed at which the spiral section advances along the fiber. This speed is determined by the fiber connection itself, independent of the observer's motion in the base space — just as the propagation speed of a wave on a grid is determined by the grid constant, not by how fast the observer runs on the grid.

11.3.2 Lorentz Transformation

The Lorentz transformation is the core of special relativity. In the fractal framework, it is the relative scaling of the grid seen by the observer when the observer moves at velocity v relative to the fiber grid.

Time dilation and length contraction are not bizarre properties of spacetime itself, but the projection shortening of the grid step in the observation direction when the observer traverses the fractal grid.

11.3.3 Einstein Field Equation and Gravity

The Einstein field equation says gravity = spacetime curvature. In Chapter 7 of this theory, gravity is a transverse closed topological ring channel with holonomy $\mathrm{Hol} = 1$.

Correspondence: General relativity is the effective description of the gravity closed channel in the smooth large-scale limit. The curvature tensor $G_{\mu\nu}$ in the Einstein equation corresponds to

the local curvature average of the fractal fiber bundle at large scales; the energy-momentum tensor $T_{\mu\nu}$corresponds to the effective mass source produced by manifold intersection (Chapter 3 A_{eff}).

General relativity's success in large-scale galactic motion, light deflection, and gravitational lensing proves that the gravity closed channel holonomy structure is correct; but it cannot explain quantum gravity because it smooths out the fractal grid.

11.4 Why the Quantum Gravity Puzzle Naturally Dissolves

The greatest theoretical difficulty of modern physics: quantum mechanics and general relativity are incompatible, requiring quantum gravity.

V4 position: This difficulty itself arises because two effective theories are mistaken for fundamental theories.

Quantum mechanics is a small-scale coarse-graining approximation, and general relativity is a large-scale smooth approximation. They are inherently projections of the same set of fractal fiber bundles in different limits, and naturally fail at intermediate scales (the quantum gravity scale) — because there neither coarse-graining nor smoothing can be applied.

This theory provides the exact geometry at intermediate scales: fiber bundle + generalized Chern number + holonomy + shadow dynamics. Quantum gravity need not be reinvented; it is the fractal geometry itself.

11.5 Correspondence Table

Modern physics concept	Corresponding object in this theory	Approximation condition
Wave function ψ	Fiber phase field	Coarse-grained statistical average
Planck constant $\hbar$	Quantization unit of the finest step of the fractal grid	Discrete grid fundamental constant
Speed of light c	Topological speed of spiral section advance along the fiber	Determined by the fiber connection
Lorentz transformation	Grid projection scaling during observer relative motion	Large-scale smooth
Spacetime curvature	Local curvature average of the fiber bundle	Large-scale smooth
Quantum entanglement	Fiber-level shared topological connection	Base-space distance far, fiber space near

Modern physics concept	Corresponding object in this theory	Approximation condition
Uncertainty principle	Geometric lower bound of the discrete fractal grid	Grid cannot be further divided
Wave-particle duality	Particle nature of the vortex standing-wave knot + wave nature of spiral propagation	Two sides of the same vortex

11.6 Chapter Summary

- Quantum mechanics and relativity are not parallel systems competing with this theory, but effective approximations of the fractal fiber bundle in different scale limits.
- Quantum mechanics = small-scale coarse-graining: the wave function is the statistical average of the fiber phase field, the uncertainty principle is the geometric lower bound of the discrete grid, entanglement is the fiber topological connection.
- Relativity = large-scale smoothing: the speed of light is the fiber topological propagation speed, the Lorentz transformation is the grid projection scaling, spacetime curvature is the fiber-bundle curvature average.
- The quantum gravity puzzle exists because two effective theories are mistaken for fundamental theories; the fractal fiber bundle itself is the exact geometry at intermediate scales.
- This chapter completes the docking of this theory with the two pillars of modern physics, neither overturning verified experiments nor failing to provide a deeper geometric unification foundation.

One-sentence summary: Quantum mechanics is the coarse-grained statistics of the fractal grid, and relativity is the smooth projection of the fractal grid; this theory is the exact geometry of the grid itself, and the two naturally emerge in their respective limits.

Chapter 12 Stellar Generator Cosmology

12.0 Chapter Positioning

This chapter is the core cosmology chapter of the entire book, constructing the V4-exclusive stellar generator cosmology model and correcting the limitations of V3's cosmological model.

The previous chapters have completed: microscopic particle topological ontology, four-force shadow dynamics, atomic and molecular topological structures, spin chirality, and quantum and relativity as fractal limit approximations. This chapter extends the scale to stars, galaxies, and the universe as a whole, answering the following key questions:

16. The universe's energy does not come from a one-time single-point big bang; what is the topological carrier of the energy source?

17. Is a star merely a furnace burning hydrogen, or a secondary vortex generator of the universe's underlying topology?
18. The physical source of dark energy and the universe's continuous background pressure;
19. Is the universe as a whole a one-way decaying evolution, or a dynamic steady-state system of global source + countless stellar secondary generators + boundary topological turn-back.

V4 core tasks for this chapter:

18. Define the stellar topological generator: a star is a secondary vortex emission source aggregate activated by the global Planck source;
19. Describe the stellar internal topological channel: the stellar core opens the fractal longitudinal vortex tube, continuously ejecting energy-substance and vortices;
20. Explain galactic structure: gravitational shadow-dynamics coupling between stellar generators, forming galaxies and galaxy clusters;
21. Provide topological interpretation of dark energy and cosmic background pressure; distinguish this model from the standard ΛCDM cosmological model;
22. Discuss the cosmic boundary: finite, boundaryless, satisfying the holonomy turn-back condition.

Inherited preliminary symbols: Φ golden ratio; $W[K] = 1/\Phi^K$; $\mathrm{Hol}(\gamma) = \mathcal{P}\exp(-\oint_\gamma \omega)$ loop holonomy; $F = P_0^{(\mathrm{local})} \cdot \eta \cdot W[K] \cdot R \cdot A_{\mathrm{eff}}$ shadow-dynamics formula.

12.1 Stars as Topological Secondary Generators

12.1.1 Generator Definition

There exists a global Planck source ($n = 0$ hierarchy) at the universe's underlying level, supplying the baseline energy of the entire fractal fiber bundle. A star does not rely purely on nuclear reactions to release stored chemical/nuclear energy; the enormous pressure at the stellar core opens a cross-hierarchy longitudinal vortex tube, becoming a secondary topological generator.

V4 core proposition:

Stellar nuclear reactions are a concomitant phenomenon of the secondary generator's operation, not the sole source of all stellar energy. The essence of a star is: a vortex generator aggregate activated by the global vacuum pressure, opening a longitudinal vortex tube, and continuously extracting topological energy-substance from the underlying fractal hierarchy to eject outward.

- When the stellar mass is large enough, the core pressure reaches the topological channel opening threshold, and the longitudinal umbilical-cord vortex tube is opened;
- The vortex transmits upward from the underlying fractal hierarchy; part converts into particles and photons radiated outward (starlight); part remains inside the star to drive nuclear fusion;

- Stellar death (white dwarf, neutron star, black hole): the core pressure changes, the longitudinal vortex tube closes, and the secondary generator stops working. A black hole is not a gravitational singularity, but a high-manifold-intersection dense region after the vortex tube has closed.

12.1.2 Vortex-Tube Characteristics of the Stellar Generator

The vortex tube corresponding to a star belongs to the longitudinal open channel, of the same topological pipe class as the strong and electromagnetic forces:

- Holonomy $\mathrm{Hol}(\gamma) \neq 1$, with cross-hierarchy energy transmission;
- Energy transmits upward along the fractal hierarchy, obeying fractal-hierarchy damping;
- Each independent star is an independent secondary emission source; the universe has massive distributed generators, not a single initial explosion source.

12.2 Galaxy Formation: Gravitational Shadow-Dynamics Coupling Between Stellar Generators

Countless stellar generators are scattered in the fractal base space; the interactions between stars are governed by Chapter 7's shadow dynamics.

10. Stars possess effective cross-sections A_{eff} produced by manifold intersection, generating gravitational shadow obstruction effects;
11. Stellar generators continuously eject vortex fields outward; the vortex fields overlap each other, forming a large-scale continuous vacuum pressure field $P_0^{(\mathrm{local})}$;
12. The long-range attraction of gravitational shadows gathers large numbers of stellar generators, forming galaxies; galaxies continue to couple, forming galaxy clusters and superclusters.

Topological explanation of galaxy rotation curves: The additional gravitational effects observed at galactic outskirts are not contributed solely by invisible dark matter particles. Part comes from the gravitational projection of the higher-dimensional submanifolds of the cosmic web (adjacent-dimensional projection effect), superposed with the local pressure correction brought by the galaxy's overall vortex field. This distinguishes itself from the standard model's explanation that introduces only dark matter particles.

12.3 Dark Energy: The Global Background Pressure of Superposed Stellar Generators

In the ΛCDM model, dark energy is an unknown negative-pressure component driving the accelerated expansion of the universe.

In the stellar generator cosmology framework: all stellar secondary generators continuously eject vortices outward; the vortex fields of countless generators superpose, forming the

universe-wide static background vacuum pressure, i.e., the observational effect corresponding to dark energy.

- The background pressure comes from the continuous supply of distributed stellar generators;
- The pressure uniformly fills the entire fractal base space, manifesting as a repulsive effect on large scales;
- Cosmological redshift (Chapter 10) is a fractal-hierarchy scaling projection effect; the observed cosmic accelerated expansion is not spacetime itself stretching, but the apparent effect brought by fractal-hierarchy projection, not true spacetime expansion.

Key distinction:

- Standard model: the universe originates from a single-point big bang; dark energy is an intrinsic property of spacetime itself.
- V4 stellar generator model: the universe has a global Planck source at the bottom + massive distributed stellar secondary generators at the top; the background pressure comes from stellar vortex ejection; the apparent expansion is a fractal projection.

12.4 Overall Topological Structure of the Universe: Finite Without Boundary, Holonomy Turn-Back

The base space of this universe is a finite, boundaryless closed topological manifold.

Any sufficiently long geodesic, traversed once around a loop, satisfies the global holonomy turn-back condition:

$$\mathrm{Hol}_{\mathrm{global}}(\gamma) = 1$$

- There is no absolutely meaningful cosmic outer boundary; there is no "outside the universe";
- Energy circulates within the global manifold: the vortices ejected by the generators, after long-range propagation, re-enter the underlying fractal hierarchy via topological turn-back, forming a closed-loop circulation;
- The universe as a whole is a dynamic steady-state system, not a one-way evolution from explosion to heat death. Local regions can have stellar birth, death, and galactic mergers; but the universe's overall average energy density remains stable over long times.

12.5 Comparison of This Model with ΛCDM Standard Cosmology

Item	ΛCDM standard cosmology	V4 stellar generator cosmology
Energy origin	Single-point cosmic big bang	Planck global baseline source + distributed stellar secondary generators

Item	ΛCDM standard cosmology	V4 stellar generator cosmology
Cosmic evolution	One-way expansion, future tends to heat death	Global dynamic steady state, local stellar birth-death circulation
Dark energy	Cosmological constant of spacetime itself	Global vacuum background pressure formed by superposition of massive stellar generators
Dark matter	Unknown fundamental particle	Adjacent-dimensional cosmic web gravitational projection + local pressure correction
Cosmological redshift	Spacetime stretching, photon wavelength stretched	Apparent projection effect from fractal-hierarchy scaling
Cosmic boundary	Infinite or finite expansion boundary	Finite boundaryless closed manifold, global holonomy turn-back

This model is compatible with all confirmed observational facts of the standard model (cosmic microwave background, galaxy distribution, gravitational lensing, etc.); it only provides a completely different topological explanation for the underlying physical mechanism, without denying the observational data.

12.6 Topological Description of the Stellar Life Cycle

- **Stellar birth:** In molecular cloud regions, boundary spheres (Chapter 8) gather in large numbers, local manifold intersection cross-sections increase, gravitational shadows contract; pressure rises, longitudinal vortex tubes open, secondary generators activate.
- **Main sequence:** The generator works stably; the vortex tube continuously transports energy-substance; nuclear fusion occurs simultaneously; stable outward radiation of photons and vortices.
- **Late-stage evolution:** Core material is consumed; internal pressure decreases; the longitudinal vortex tube gradually contracts. The vortex tube partially closes; the generator power changes; the star expands into a red giant.
- **Stellar death:** Pressure falls below the threshold; the longitudinal vortex tube completely closes; the secondary generator stops. The remaining material contracts under gravitational shadow, forming a white dwarf or neutron star; if the mass is large enough, a black hole forms (dense manifold-intersection region, vortex tube completely closed).

12.7 Chapter Summary

- A star is not merely a nuclear reaction furnace, but a secondary topological generator activated by the global Planck source; the stellar core opens the longitudinal vortex tube, continuously extracting energy-substance from the underlying fractal hierarchy to eject outward; nuclear reactions are a concomitant effect of the generator's operation.

- Galaxies are gathered by gravitational shadow-dynamics coupling between stellar generators; the additional gravitational effect of galaxies partially comes from the higher-dimensional submanifold projection of the cosmic web.
- The observed dark energy effect comes from the global vacuum background pressure formed by the continuous vortex ejection superposition of countless stellar generators in the universe; the apparent accelerated expansion of the universe is a fractal-hierarchy projection effect.
- The entire universe is a finite boundaryless closed topological manifold, globally satisfying the holonomy turn-back, with energy circulation and dynamic steady state; local stellar birth-death and galactic evolution continue to occur, but there is no one-way evolution path from big bang to heat death.
- This chapter completes the theoretical chain from microscopic particles, atoms and molecules, stellar generators to the entire universe, incorporating cosmology into the unified fractal fiber-bundle framework.

One-sentence summary: The universe's energy supply comes from the underlying Planck global source and massive distributed stellar secondary generators; the apparent cosmic expansion is a fractal projection effect; the universe is a finite boundaryless, energy-circulating, dynamic steady-state topological manifold.

Chapter 13 Theoretical Predictions, Observable Tests, and Distinguishing Experiments

13.0 Chapter Positioning

This chapter is the validation chapter of the entire book. The previous twelve chapters have built a complete fractal fiber-bundle ontology, microscopic particles, atoms and molecules, spin chirality, four-force shadow dynamics, quantum and relativity limits, and stellar generator cosmology. Any physical theory must possess observable predictions and falsifiable conditions. This chapter centrally sorts V4 theory's exclusive predictions and designs observable and experimental schemes that can distinguish this theory from mainstream standard models (particle physics Standard Model, ΛCDM cosmology).

Core tasks of this chapter:

1. Distinguish compatible predictions (consistent with existing model conclusions, non-conflicting) from exclusive distinguishing predictions (V4-exclusive, usable for experimental adjudication);
2. Divide into microscopic experiments and astronomical observations, listing test schemes;
3. Discuss the observational indicators and measurement difficulties corresponding to each prediction;

4. Analyze the theory's falsification criteria: which experimental results would directly falsify V4's stellar generator and fractal fiber-bundle framework;
5. Analyze how currently observed phenomena are self-consistently explained by this theory;
6. Pave the way for Chapter 14's theoretical summary.

Inherited preliminary symbols: Φ golden ratio; $W[K] = 1/\Phi^K$; $\mathrm{Hol}(\gamma) = \mathcal{P}\exp(-\oint_\gamma \omega)$ loop holonomy; $F = P_0^{(\mathrm{local})} \cdot \eta \cdot W[K] \cdot R \cdot A_{\mathrm{eff}}$ shadow-dynamics formula.

13.1 Two Types of Prediction Definitions

Compatible prediction: The conclusion holds in both the standard model and V4 theory; cannot be used to distinguish the two theories. Example: hydrogen atomic energy levels, covalent bond energies, gravitational deflection of light near the Sun. Such phenomena have already been experimentally verified in large quantities; V4 theory can reproduce the observational results within the fractal framework, but cannot serve as exclusive evidence of this theory.

Distinguishing prediction (core): A result predicted by V4 theory that differs significantly from the standard model's expectation. Once high-precision observation confirms this phenomenon, it provides support for this theory; if observation denies it, the corresponding part of the framework must be modified or even falsified.

13.2 Microscopic-Field Testable Predictions

13.2.1 Small Correction Effect Associated with Spin-Fiber Winding

In standard quantum mechanics, spin exists only as an intrinsic quantum number; there is no underlying continuous topological structure of fiber winding.

V4 prediction: In extremely high-precision spin precession experiments, there exists an extremely weak nonlinear correction term, arising from the coupling of the fiber winding number with the generalized Chern number K.

1. Experimental scheme: ultra-high-precision electron/proton spin precession precision measurement (similar to the g−2 experiment);
2. Expectation: in addition to known quantum electrodynamic corrections, there exists an additional tiny offset;
3. Falsification condition: at higher precision, this additional offset remains zero continuously.

13.2.2 Topological Pattern of Chiral Molecular Tiny Parity Energy Difference

The standard model holds that the tiny energy difference of molecular chirality comes only from the tiny coupling of the weak interaction.

V4 prediction: The energy difference of chiral isomers has a fittable functional relationship with the topological volume of the molecular boundary sphere and the shared K_{shared}; the energy difference exhibits a fractal scaling pattern with the topological complexity of the molecular space.

- Experimental scheme: high-precision spectroscopic measurement of the energy-level difference of chiral molecular pairs, compared with molecular topological structure;
- Falsification condition: the energy difference does not exhibit the expected scaling relation with the boundary-sphere topological parameters.

13.2.3 Fiber Topological Damping of Quantum Entanglement Propagation

Standard quantum mechanics holds that entanglement correlation does not decay with distance (no signal transmission).

V4 prediction: Entanglement relies on fiber topological connection; when the base-space distance is very large, the fiber vortex experiences fractal-hierarchy damping, and the entanglement fidelity exhibits a slow decay related to the fractal hierarchy, not purely ideal non-decay.

- Experimental scheme: long-distance entangled photon distribution, measurement of entanglement fidelity across a hundred-kilometer-level quantum channel;
- Note: the effect is extremely weak, requiring extremely high signal-to-noise ratio;
- Falsification condition: long-distance entanglement fidelity completely lacks the expected fractal damping decay.

13.3 Astronomical and Cosmological Distinguishing Predictions (V4 Core Tests)

13.3.1 Stellar Generator: A Tiny Extra Flux in Stellar Energy Beyond Pure Nuclear Fusion

ΛCDM and standard stellar models: all stellar radiant energy comes uniquely from nuclear fusion.

V4 prediction: The stellar core longitudinal vortex tube extracts a small amount of energy-substance from the underlying fractal hierarchy, so the total stellar radiant flux has a tiny excess radiation beyond the value calculated by nuclear fusion theory. This excess flux varies with the stellar core pressure (mass): the larger the stellar mass, the higher the vortex tube opening degree, and the higher the excess ratio.

- Observational scheme: high-precision photometry and spectroscopy, fine energy-budget accounting for massive stars, subtracting all nuclear reaction energy production;

- Falsification condition: extremely high precision energy measurement confirms that stellar radiation is exactly equal to nuclear fusion energy production, with no excess flux whatsoever.

13.3.2 Scaling Law of Galactic Gravitational Anomaly (Dark Matter Effect) with Galactic Topological Morphology

Standard model: galactic gravitational anomaly is dominated by dark matter particles, correlated only with total mass.

V4 prediction: The extra gravitational contribution of galaxy rotation curves comes from adjacent-dimensional cosmic web gravitational projection; the strength of this effect depends on the topological structure of the galaxy's overall vortex field, not purely on the total baryonic mass of the galaxy. Galaxies of the same mass but different morphology exhibit fractal scaling differences in gravitational anomaly amplitude.

- Observational scheme: large-scale galaxy surveys, statistics of rotation curve deviations among galaxies of equal mass but different configuration;
- Falsification condition: gravitational anomaly strictly proportional only to galactic baryonic mass, unrelated to galactic topological morphology.

13.3.3 Cosmological Redshift Not Following Standard Expansion Evolution with Distance

ΛCDM: redshift z with distance is determined by spacetime expansion, conforming to FLRW metric evolution.

V4 prediction: Redshift is a fractal-hierarchy scaling projection effect; the redshift-distance relation will deviate from the high-order corrections of the standard expansion model; there is no true acceleration from spacetime stretching. In high-redshift celestial observations, high-order deviations from ΛCDM predictions can be found.

- Observational scheme: high-redshift supernovae and gravitational-wave standard sirens for high-precision ranging;
- Falsification condition: larger-sample high-redshift observations strictly conform to the ΛCDM expansion curve, with no systematic deviation from fractal projection.

13.3.4 Global Background Pressure (Dark Energy) Correlated with Stellar Number Density

ΛCDM: dark energy is a cosmological constant, constant throughout space, unrelated to stellar distribution.

V4 prediction: Background vacuum pressure is generated by the superposition of stellar generator vortices; in cosmic regions with higher stellar density, the local background pressure has a tiny increase, bringing a detectable local repulsive effect.

- Observational scheme: comparing large-scale gravitational lensing in stellar-dense regions and void regions;
- Falsification condition: background pressure completely uniform, with no correlation with stellar distribution.

13.4 Self-Consistent Explanation of Already-Observed Phenomena (Compatible Part, Non-Falsifiable)

These phenomena are not newly predicted by V4; only underlying topological explanations are provided, and they cannot be used to distinguish models:

- Cosmic microwave background radiation: global vortex thermal equilibrium radiation of the stellar generator system;
- Light gravitational deflection and gravitational lensing: large-scale smooth limit of the gravity closed channel ($\mathrm{Hol} = 1$), general relativity is effectively approximate;
- Atomic spectra and chemical bond energies: double-vortex and boundary-sphere topological resonance;
- Weak interaction parity non-conservation: dynamical chirality of the weak-force connection itself.

Being able to explain existing observations does not prove the theory correct; the persuasive power of a physical theory comes from experimental verification of distinguishing predictions.

13.5 Summary of Falsification Criteria

Any one of the following observational results, if established, would directly falsify the V4 core framework:

- High-precision energy budgets of massive stars show absolutely no extra radiation flux beyond nuclear fusion; the stellar generator hypothesis is untenable.
- Long-distance entanglement fidelity shows no fractal damping decay whatsoever, fully consistent with ideal quantum mechanics.
- Galactic gravitational anomaly strictly proportional only to baryonic mass, unrelated to galactic vortex topological morphology.
- High-redshift celestial ranging strictly satisfies the ΛCDM spacetime expansion curve, with no systematic deviation from fractal projection.
- Cosmic background pressure in stellar-dense regions and cosmic voids is completely identical, not varying with stellar number density.

13.6 Practical Difficulties of Experiments and Observations

13. The effects of most exclusive predictions are extremely weak, requiring next-generation higher-precision equipment (next-generation gravitational-wave detectors, ultra-high-precision stellar photometry, long-baseline quantum optical experiments, large-field-of-view galaxy surveys); the precision of existing instruments is insufficient to resolve tiny deviations.
14. Astronomical observations have large amounts of confounding interference (dust obscuration, galactic internal dynamics, instrumental systematic errors), requiring complex data cleaning.
15. The correction terms in microscopic precision experiments are extremely small and are easily masked by known quantum corrections.

13.7 Chapter Summary

- Theoretical predictions are divided into compatible predictions and distinguishing predictions; only the latter can be used to test and falsify this theory.
- Microscopic-direction predictions: tiny topological correction of spin precession, fractal scaling of chiral molecular parity energy difference, and long-distance entanglement fiber damping decay.
- Cosmological core predictions: stars have a tiny excess radiation beyond nuclear fusion; galactic gravitational anomaly depends on topological morphology; redshift is a fractal projection; background pressure correlates with stellar number density.
- The falsification conditions of V4 theory are explicitly listed; if the corresponding observational results appear, the core assumptions of the theory must be abandoned or modified.
- Existing observational phenomena can be self-consistently explained within this framework, but cannot serve as evidence; the final adjudication of the theory depends on future high-precision experiments and astronomical observations.

One-sentence summary: V4 fractal unified field theory provides a series of observable exclusive predictions; future precision microscopic experiments and astronomical observations can determine whether the fractal fiber bundle and stellar generator hypotheses hold.

Chapter 14 Summary and Outlook

14.0 Chapter Positioning

This chapter is the summary chapter of the entire book. The previous thirteen chapters have completed the full theoretical construction, from cosmic ontology, particle topology, four-force dynamics, atomic and molecular structure, spin chirality, fractal limits, and stellar generator cosmology, to observable tests. This chapter summarizes the core conclusions of the entire book and explains the theory's positioning and future research directions.

14.1 Core Conclusions of the Entire Book

Conclusion One: The universe is a fractal fiber bundle.

The base space is the layer axis B, with each layer amplified by 24 times; the fiber is the phase circle S^1; the total space $E = B \times F$ constitutes a trivial fiber bundle. Spiral motion is equivalent to a fiber-bundle section; the particle is the topological jamming of the spiral wave on the Luoshu grid.

Conclusion Two: Center Point 5 is the unique zero point of the base space.

Global Center Point 5 is located at $n = 0$ (Planck scale). The fractal Center Point 5 at each scale is the local trivialization projection of the global zero point at the corresponding scale, satisfying the nested relation $C_0 \subset C_1 \subset C_2 \cdots$. Containment does not equal revolution.

Conclusion Three: K is the generalized first Chern number, and simultaneously the resonance condition.

K is both the fiber-bundle topological distortion measure and the phase-matching criterion. Shared K is equivalent to local fiber-bundle isomorphism, and is the topological foundation of chemical bonds and particle family resonance.

Conclusion Four: Particle mass originates from manifold transverse intersection.

Particle rest mass is a topological invariant of submanifold transverse intersection, $M \propto I \cdot M_0$. The 33-particle mass spectrum has an overall average deviation of 0.89%; all V3 numerical results are retained.

Conclusion Five: The four fundamental forces are unified in topological shadow dynamics.

The differences among the four fundamental forces are differences in topological structure, not differences in spacetime and fields. Gravitational weakness originates from the scarcity of topological degrees of freedom (ratio 2.03%); its channel is the universe's only perfectly lossless transverse closed fractal ring (holonomy $\mathrm{Hol} = 1$). The Φ^5 exponential decay of the longitudinal umbilical-cord channel applies only to the strong and electromagnetic forces; V3's old gravitational decay model is discarded.

Conclusion Six: Atomic orbits are phase-matching surfaces of double vortices.

The electron orbit is the phase-matching surface between the Planck vortex and the atomic-nucleus vortex; the phase condition is $\Delta\phi = \pm\pi/2$. The Pauli exclusion principle comes from the same-orbit two-electron phase anti-phase condition $\Delta\phi_{ee} = \pi$.

Conclusion Seven: Chemical bonds are shared-K topological resonance within the boundary sphere.

Two atomic vortices realize a shared generalized Chern number K_{shared} topological resonance bound state inside the boundary sphere; the bond energy is jointly determined by the boundary-sphere topological weight and the intersection cross-section.

Conclusion Eight: Quantum mechanics and relativity are limit approximations of the fractal fiber bundle.

Quantum mechanics = small-scale coarse-graining; relativity = large-scale smoothing. The quantum gravity puzzle exists because two effective theories are mistaken for fundamental theories.

Conclusion Nine: The universe is a finite boundaryless, energy-circulating, dynamic steady state.

A star is not a small white hole; it is an aggregate of countless locally activated secondary vortex sources; the universe is not a pile of ashes after a one-time big bang, but a dynamic steady state of global source + distributed stellar secondary generators + boundary topological turn-back.

Conclusion Ten: The theory provides falsifiable exclusive predictions.

Stellar excess radiation, galactic gravitational anomaly depending on topological morphology, redshift as fractal projection, background pressure correlated with stellar number density, tiny topological spin correction, entanglement fiber damping decay, etc.

14.2 Theory Positioning

V4 positioning: **geometric reconstruction, not paradigm overthrow.**

1. All experimental comparison results of V1–V3 that have passed numerical verification are retained;
2. The work of V4 is to elevate V3's phenomenological and algebraic models to the modern mathematical physics framework of fiber bundles- differential geometry- topology;
3. V4 corrected three theoretical gaps of V3: the ontology of Center Point 5, the conceptual ambiguity of K, and the gravitational channel decay factor.

V4 provides a deeper geometric ontology, not an overthrow of the standard model and ΛCDM. The two naturally coexist in their respective applicable limits.

14.3 Future Research Directions

1. **Strict discretization-continuum mapping**: strictly analytically deriving the Luoshu discrete sampling amplification factor 24 from the symplectic-manifold constraint conditions;
2. **Strict derivation of K splitting coefficient 2.164**: deriving the V3 empirical coefficient from the first principles of the second generalized Chern number;
3. **Complete topological channel of the weak force and holonomy conditions**: establishing an explicit expression for the weak-force chiral connection;
4. **First-principles derivation of the light-matrix response function R**: deriving the response function from the first principles of vortex theory;
5. **Topological formulas for complex polyatomic molecules, conjugated systems, and hydrogen bonds**: constructing multi-center shared K network models;
6. **Fractal correspondence of quantum field theory renormalization group**: studying the operation of QED and QCD renormalization in the fractal framework;

7. **Advancement of experimental tests**: high-precision photometry of stellar excess radiation, large-field-of-view galaxy surveys, long-baseline quantum optical experiments, next-generation gravitational-wave detectors.

14.4 One-Sentence Summary

The universe is a layer-by-layer nested fractal fiber bundle; Center Point 5 is the unique zero point of the base space; K is the generalized Chern number and resonance condition; particle mass is manifold intersection; the four forces are topological shadow dynamics; quantum mechanics and relativity are fractal limits; the universe is a finite boundaryless, energy-circulating, dynamic steady state.

Image first, formula follows; geometric reconstruction, not paradigm overthrow.

Appendix

Appendix A 33 Particle Mass Calculation Formulas and Predictions

A.1 Lepton Mass

Lepton mass formula:

$$M = M_0 \times \Phi^k$$

where $M_0 = 0.8255_{\text{MeV}}$, Φ is the golden ratio, and $k = 1 - 5\Delta n - \delta$.

Electron

$\Delta n = 0, \delta = 0, k = 1.$

$$M_e = 0.8255 \times \Phi^1 = 0.5102_{\text{MeV}}$$

Experimental value 0.511 MeV, deviation -0.16%.

Muon

$\Delta n = 2.24, \delta = 0, k = -10.2.$

$$M_\mu = 0.8255 \times \Phi^{-10.2} = 111.9_{\text{MeV}}$$

Experimental value 105.7 MeV, deviation $+5.85\%$.

Tau

$\Delta n = 3.15 > 3$, deep-layer locking triggered, $\delta = 2\Phi = 1.236, k = -15.986.$

$$M_\tau = 0.8255 \times \Phi^{-15.986} = 1808_{\text{MeV}}$$

Experimental value 1777 MeV, deviation $+1.75\%$.

A.2 Quark Mass

Quark mass formula:

$$M_q = M_{\text{base}} \times 24^{I_q}$$

Quark	I_q	Theoretical value (MeV)	Experimental value (MeV)	Deviation
u	0.4691	2.27	2.30	−1.3%
d	0.6976	4.69	4.70	−0.2%
s	−0.0309	95.8	95.0	+0.8%
c	0.7809	1265	1270	−0.4%
b	0.2691	4179.6	4180	−0.01%
t	1.4451	175.6 GeV	173.0 GeV	+1.5%

A.3 Gauge Boson Mass

$M = M_b \times \Phi^{-5} \times f$, $M_b = 4179.6_{\text{MeV}}$.

Particle	f	Theoretical value (GeV)	Experimental value (GeV)	Deviation
W	$\sqrt{3}$	80.3	80.4	−0.1%
Z	$\sqrt{3} \times \sqrt{107/83}$	91.1	91.2	−0.1%
Higgs	$\sqrt{3} \times \pi/2$	126.1	125.1	+0.7%

A.4 Composite Particle Mass

Particle	Theoretical value (MeV)	Experimental value (MeV)	Deviation
π meson	140.96	139.6	<2%
K^+	500.1	493.7	+1.3%
J/ψ	3097	3096.9	~0%
Υ	9467.2	9460.3	~0%

Particle	Theoretical value (MeV)	Experimental value (MeV)	Deviation
D^0	1866.27	1864.8	+0.1%
Proton	938.23	938.3	−0.01%
Neutron	939.6	939.6	~0%
Ω^-	1672.4	1672.5	~0%

A.5 Overall Statistics of 33 Particles

Category	Number	Average deviation	Maximum deviation
Leptons	3	2.59%	5.85%
Quarks	6	0.76%	1.50%
Gauge bosons	3	0.28%	0.70%
Composite particles	21	0.52%	1.90%
Total	33	0.89%	5.85%

A.6 New Particle Predictions

A.6.1 Fourth-Generation Lepton

Layer number $n_4 = 13.08, \Delta n = 3.71, \delta = 1.675, k = -19.225$.

$$M_4 = 0.8255 \times \Phi^{-19.225} = 8.60_{\text{GeV}}$$

Charge same as electron, muon, tau (−1), spin 1/2. Located in the deep-layer locking region; decay mode involves yin-side energy return flow, manifested in detectors as large missing transverse momentum plus an isolated lepton.

Search path: Belle II screening back-to-back lepton pairs in the 10.58 GeV dataset, scanning 8.5–8.7 GeV.

A.6.2 Fifth-Generation Lepton

Layer number $n_5 = 12.73, \Delta n = 4.06, \delta = 1.891, k = -21.191$.

$$M_5 = 0.8255 \times \Phi^{-21.191} = 22.15_{\text{GeV}}$$

A.6.3 Sixth-Generation Lepton

Layer number $n_6 = 12.51, \Delta n = 4.28, \delta = 2.027, k = -22.427.$

$$M_6 = 0.8255 \times \Phi^{-22.427} = 40.16_{\text{GeV}}$$

A.6.4 Fourth-Generation Quarks

Using the fourth-generation lepton 8.60 GeV as baseline:

Quark	I_q	Mass
u_4	0.4691	38.2 GeV
d_4	0.6976	78.9 GeV
s_4	−0.0309	7.79 GeV
c_4	0.7809	102.9 GeV
b_4	0.2691	20.2 GeV
t_4	1.4451	849.5 GeV

A.6.5 Fifth-Generation Quarks

Using the fifth-generation lepton 22.15 GeV as baseline:

Quark	I_q	Mass
u_5	0.4691	98.3 GeV
d_5	0.6976	203.3 GeV
s_5	−0.0309	20.1 GeV
c_5	0.7809	265.1 GeV
b_5	0.2691	52.1 GeV
t_5	1.4451	2.19 TeV

A.6.6 Sixth-Generation Quarks

Using the sixth-generation lepton 40.16 GeV as baseline:

Quark	I_q	Mass
u_6	0.4691	178.3 GeV

Quark	I_q	Mass
d_6	0.6976	368.6 GeV
s_6	−0.0309	36.4 GeV
c_6	0.7809	480.6 GeV
b_6	0.2691	94.4 GeV
t_6	1.4451	3.97 TeV

A.6.7 Collider Search Paths

Fourth-generation lepton (8.60 GeV):

- Belle II: screening back-to-back lepton pairs in the 10.58 GeV dataset, scanning 8.5–8.7 GeV;
- LHCb: reconstructing the ϕ meson, searching for a 7.7–7.8 GeV long-lived resonance (corresponding to the fourth-generation s-type quark);
- LEP data reanalysis: retrieving public data, separately analyzing the 10–25 GeV dilepton invariant mass.

Fifth-generation lepton (22.15 GeV):

- LEP 91.2 GeV Z resonance dataset, decay channel $Z \to \tau_5^+ \tau_5^-$.

Sixth-generation lepton (40.16 GeV):

- Future colliders (ILC, FCC-ee).

A.6.8 Image Sources of the Mass Formulas

Lepton mass formula $M = M_0 \Phi^k$ source: the image of the spiral jammed by the grid; see V1 Volume 1, V3 Chapters 1–3.

Quark mass formula $M_q = M_{\text{base}} \times 24^{I_q}$ source: the transverse spring image; see V1 Volume 8 Chapter 4, V3 Chapter 3.

Gauge boson mass formula $M = M_b \Phi^{-5} f$ source: the pipe resonance image; see V1 Volume 4, V3 Chapter 6.

Composite particle mass formula $M = M_{\text{quark sum}} + E_{\text{binding}}$ source: the boundary-sphere image; see V1 Volume 9 Chapter 4, V3 Chapter 7.

Appendix B Derivation of the Electron Layer Number 16.79

Electron Compton wavelength:

$$\lambda_e = \frac{h}{m_e c} = 2.426 \times 10^{-12}\ \text{m}$$

Planck length $\ell_P = 1.616 \times 10^{-35}$ m.

$$n_e = \log_{24}\left(\frac{\lambda_e}{\ell_P}\right) = \log_{24}(1.501 \times 10^{23})$$

$$\ln(1.501 \times 10^{23}) = 53.365,\ \ln 24 = 3.178.$$

$$n_e = 53.365/3.178 = 16.79$$

Detailed derivation see V3 Chapter 1, V1 Volume 3.

Appendix C Complete Gravity Calculation

C.1 Earth-Sun Gravity

System parameters

Sun layer number $n_{\text{Sun}} = 60.5$, Earth layer number $n_{\text{Earth}} = 56.5$, layer-number difference $\Delta n = 4$.

Sun interception rate

$$\eta_{\text{Sun}} = 89.87\% + 8.10\% = 97.97\%$$

Earth-layer background pressure

$P_0^{(\text{Earth})} = 3.54 \times 10^{11}$ Pa (inferred from the measured Earth-core pressure)

Earth effective cross-section

$$A_{\text{eff}} = \pi R_{\text{Earth}}^2 \times \frac{1}{\Phi^5} = 1.275 \times 10^{14} \times 11.09 = 1.414 \times 10^{15}\ \text{m}^2$$

Cross-layer decay factor

$$(\Phi^5)^4 = 6.61 \times 10^{-5}$$

Shadow formula calculation

$$F = 3.54 \times 10^{11} \times 0.9797 \times 6.61 \times 10^{-5} \times 1.414 \times 10^{15} = 3.24 \times 10^{22}\ \text{N}$$

Comparison with Newtonian value

$$F_N = G\frac{M_{\text{Sun}}M_{\text{Earth}}}{r^2} = 3.54 \times 10^{22}\ \text{N}$$

Deviation -8.5%.

Detailed derivation see V1 Volume 10 Chapter 4, V3 Chapter 7.

C.2 Coulomb Force

System parameters

Electron layer $n_e = 16.79$, proton spatial layer $n_p = 14.42$, layer-number difference $\Delta n = 2.37$.

Proton interception rate

$$\eta_p = \frac{2.025\%}{24^4} = 6.10 \times 10^{-8}$$

Electron-layer background pressure

$$P_0^{(\text{electron})} = 4.09 \times 10^{21}\ \text{Pa}$$

Cross-layer decay factor

$$(\Phi^5)^{2.37} = 3.33 \times 10^{-3}$$

Electron effective cross-section

$$A_{\text{eff}}(e) = \pi a_0^2 \times \frac{1}{\Phi^5} = 8.80 \times 10^{-21} \times 11.09 = 9.76 \times 10^{-20}\ \text{m}^2$$

Shadow formula calculation

$$F = 4.09 \times 10^{21} \times 6.10 \times 10^{-8} \times 3.33 \times 10^{-3} \times 9.76 \times 10^{-20} = 8.11 \times 10^{-8}\ \text{N}$$

Comparison with Coulomb force

$$F_{\text{Coulomb}} = \frac{e^2}{4\pi\epsilon_0 a_0^2} = 8.24 \times 10^{-8}\ \text{N}$$

Deviation -1.6%.

Detailed derivation see V1 Volume 10 Chapter 5, V3 Chapter 7.

C.3 Proton-Proton Gravity (Detailed Derivation)

（1）System parameters

- Proton layer number $n_p = 19.16$
- Proton Compton wavelength $\lambda_p = h/(m_p c) = 1.321 \times 10^{-15}\ \text{m}$
- Proton Compton volume $V_p = \lambda_p^3 = 2.305 \times 10^{-45}\ \text{m}^3$

- Proton rest energy $E_p = m_p c^2 = 1.503 \times 10^{-10}$ J

（2）Proton-layer background pressure

Energy density:

$$\rho_p = \frac{E_p}{V_p} = \frac{1.503\times10^{-10}}{2.305\times10^{-45}} = 6.52 \times 10^{34}$$ Pa

Yang-side ratio: The proton layer is 2.37 layers deeper than the electron layer; the yang-side ratio approaches 1 from the electron layer's 5/7.

$$\text{Yang-side ratio} = 1 - \left(1 - \tfrac{5}{7}\right) \times \Phi^{2.37} = 1 - 0.2857 \times 0.3197 = 0.9087$$

$$P_0^{(\text{proton})} = 6.52 \times 10^{34} \times 0.9087 = 5.92 \times 10^{34}$$ Pa

（3）Cross-layer decay factor

From the proton internal scale to the proton external scale:

$$\frac{\ell_P}{\lambda_p} = \frac{1.616\times10^{-35}}{1.321\times10^{-15}} = 1.22 \times 10^{-20}$$

$$\left(\frac{\ell_P}{\lambda_p}\right)^{1+\Phi} = (1.22 \times 10^{-20})^{1.618} = 1.39 \times 10^{-32}$$

（4）Proton interception rate

$$\eta_p = \frac{2.025\%}{24^4} = 6.10 \times 10^{-8}$$

（5）Revolution coupling correction factor

Three quarks in revolution have drag; the drag factor is $\Phi^2 = 0.382$.

（6）Local pressure

$$P_0^{(\text{local})} = 5.92 \times 10^{34} \times 1.39 \times 10^{-32} \times 6.10 \times 10^{-8} \times 0.382 = 1.92 \times 10^{-5}$$ Pa

（7）Effective cross-section

$$A_{\text{eff}} = \pi\lambda_p^2 = \pi \times (1.321 \times 10^{-15})^2 = 5.48 \times 10^{-30}$$ m²

（8）Proton-proton gravity

$$F = 1.92 \times 10^{-5} \times 5.48 \times 10^{-30} = 1.05 \times 10^{-34}$$ N

Experimental value: 1.07×10^{-34} N. **Deviation:** -1.9%.

（9）Key distinctions of the gravity formula

- $P_0^{(\text{proton})}$ is the proton internal pressure, not the pressure between two protons;

- To go from internal pressure to external pressure, one must multiply by the cross-layer decay factor $(\ell_P/\lambda_p)^{1+\Phi}$;
- Proton-proton gravity follows the transverse channel, with holonomy $\mathrm{Hol}(\gamma) = 1$, no cross-layer decay;
- V3's old model misapplied the longitudinal umbilical-cord decay factor $(\Phi^5)^{38.32}$, giving a 15-order-of-magnitude deviation; V4 discards it.

Detailed derivation see V1 Volume 10 Chapter 6, V3 Chapter 7.

Appendix D Complete Ionization Energy Calculation

D.1 Formula

$E = P_0 \times \sum_i A_i \times (\Phi^5)^{d_i}$

$P_0 = 59.2_{\text{eV}}$, anchored by lithium's first ionization energy.

D.2 Depth

- $d_{1s} = d_{\text{free}} - (\Phi + \Phi^2) = 2.37 - 1.0 = 1.37$
- $d_{2s} = d_{\text{free}} - \Phi = 2.37 - 0.618 = 1.75$
- $d_{2p} = 1.74 + \delta$
- Hydrogen: $d_H = \Phi = 0.618$

Breathing correction:

$\delta = -\Phi^{3/2} \times \frac{|h-q|}{\sqrt{1+(h-q)^2}}$

Only effective for 2P and above.

D.3 Complete Calculation for Hydrogen

Hydrogen: $1S^1$, $d_{1s} = 0.618$, $A_{1s} = 1$.

$(\Phi^5)^{0.618} = 0.09017^{0.618} = 0.226$

$E_H = 59.2 \times 1 \times 0.226 = 13.4_{\text{eV}}$

Experimental value 13.6 eV, deviation -1.5%.

D.4 Complete Calculation for Lithium

Lithium: $1S^2\ 2S^1$.

16. $d_{1s} = 1.37,\ A_{1s} = 2,\ (\Phi^5)^{1.37} = 0.0375$
17. $d_{2s} = 1.75,\ A_{2s} = 1,\ (\Phi^5)^{1.75} = 0.0150$

$$E_{Li} = 59.2 \times (2 \times 0.0375 + 1 \times 0.0150) = 59.2 \times 0.0900 = 5.32_{\text{eV}}$$

Experimental value 5.39 eV, deviation -1.3%.

D.5 Second-Period Verification

Element	Theoretical value (eV)	Experimental value (eV)	Deviation
B	8.35	8.30	+0.6%
C	11.41	11.26	+1.3%
N	14.47	14.53	−0.4%
O	14.52	13.62	+6.6%
F	16.57	17.42	−4.9%
Ne	22.66	21.57	+5.1%

Pure prime states MAD 0.8%; composite-containing states MAD 5.5%.

D.6 Third-Period Verification

Element	Theoretical value (eV)	Experimental value (eV)	Deviation
Na	5.14	5.14	0%
Mg	7.65	7.65	0%
Al	5.99	5.99	0%
Si	8.24	8.15	+1.0%
P	10.49	10.49	0%
S	10.52	10.36	+1.5%
Cl	12.03	12.97	−7.2%
Ar	16.50	15.76	+4.7%

D.7 Fourth, Fifth, Sixth Periods

Period	Number of elements	MAD
4	6	2.2%
5	6	2.7%
6	6	6.1%

D.8 Island-of-Stability Prediction

Z = 126: $E_{126} = 6.0 \pm 1.0_{\text{eV}}$.

Detailed derivation see V1 Volume 19, V3 Chapter 8.

Appendix E Molecular Bond Energy Calculation Examples

E.1 Formula

$$E_{\text{bond}} = P_0^{(\text{mol})} \times a_A \times a_B \times \Phi^{-K_{\text{shared}}}$$

$P_0^{(\text{mol})} = 3.83_{\text{eV}}$, anchored by H_2.

E.2 Complete Calculation for H_2

- Two hydrogen atoms, $a_A = a_B = 1.000$
- $K_{\text{shared}} = 0$(same period)
- $\kappa = 1.18$(hydrogen molecule pairing enhancement)

$$E_{H_2} = 3.83 \times 1.000 \times 1.000 \times 1 \times 1.18 = 4.52_{\text{eV}}$$

Experimental value 4.52 eV, deviation 0%.

E.3 Complete Calculation for HF

1. $a_A = 1.000$(H), $a_B = 0.771$(F)
2. $K_{\text{shared}} = 1$(cross-period)

$$E_{\text{HF}} = 3.83 \times 1.000 \times 0.771 \times 1.618 = 4.78_{\text{eV}}$$

Experimental value 5.90 eV, deviation -19%.

E.4 Complete Calculation for Li_2

1. $a_A = a_B = 0.616$
2. $K_{\text{shared}} = 0$

$$E_{Li_2} = 3.83 \times 0.616 \times 0.616 = 1.45_{\text{eV}}$$

Experimental value 1.06 eV, deviation $+37\%$.

E.5 Precision Summary

Bond type	Number of molecules	MAD
Pure covalent bonds	7	10.3%
Weakly polar	3	10.8%
Covalent + weakly polar	15	10.5%

Detailed derivation see V1 Volume 9, V3 Chapter 9.

Appendix F K Values of Twin Primes and the Goldbach Conjecture

F.1 Twin Primes

Twin primes $(9K+2, 9K+4)$ share the same K.

$$K = \frac{p-2}{9}$$

Simple proof

Let p and $p+2$ be twin primes, both greater than 3.

By the prime digit-root constraint, primes greater than 3 can only land on the six stations of the multiply-by-2 orbit: $\{1, 2, 4, 5, 7, 8\}$.

If the digit root of p is r, then the digit root of $p+2$ is $r+2 \mod 9$.

Requiring both r and $r+2$ to land in $\{1, 2, 4, 5, 7, 8\}$.

Enumerating digit roots: $r = 1 \to 3$ (not in); $r = 2 \to 4$ (in); $r = 4 \to 6$ (not in); $r = 5 \to 7$ (in); $r = 7 \to 9$ (not in); $r = 8 \to 1$ (in).

So the twin-prime digit-root pairs can only be $(2, 4)$, $(5, 7)$, $(8, 1)$.

Three types corresponding to K values:

- Type I: $(9K+2, 9K+4)$, e.g., 11 and 13 ($K = 1$)

- Type II: $(9K+5, 9K+7)$, e.g., 5 and 7 ($K=0$)
- Type III: $(9K+8, 9K+1)$, e.g., 17 and 19 ($K=1$)

All three types of twin primes share the same K value.

Detailed proof see V1 Volume 18, V3 Chapter 12.

F.2 Goldbach Conjecture

For any even number N, there exist two primes p_1, p_2, such that $N = p_1 + p_2$.

In the K-value framework:

$$K_{p_1} + K_{p_2} = K_N$$

Simple proof

Let N be even. Write N in the form $9K_N + M_N$, where M_N is the digit root of N.

If $p_1 = 9K_1 + M_1, p_2 = 9K_2 + M_2$ are both primes, and $p_1 + p_2 = N$, then:

$$9(K_1 + K_2) + (M_1 + M_2) = 9K_N + M_N$$

So $K_1 + K_2 + \lfloor (M_1 + M_2)/9 \rfloor = K_N$, and $(M_1 + M_2) \mod 9 = M_N$.

That is, the K values satisfy an additive relation (with possible carry correction).

Detailed proof see V1 Volume 18, V3 Chapter 12.

F.3 Laws

- Twin primes share the same K value;
- The Goldbach conjecture's K values satisfy an additive relation.

Appendix G K-Value Laws

G.1 Definition of K

K is the generalized Chern number of the fiber bundle, and simultaneously the resonance condition.

$$K \equiv \lim_{n^* \to n_{\max}} \frac{i}{2\pi} \int_{[0,n^*]} \Omega$$

G.2 K-Value Hierarchies

Hierarchy	K value	Corresponding structure
K_1	Number-theory layer	Twin primes

Hierarchy	K value	Corresponding structure
K_2	Electron layer	Electron orbit
K_3	Nucleon layer	Nuclear force
K_4	Atomic layer	Molecular bond
K_5	Molecular layer	Protein
K_6	Cellular layer	Human body
K_n	Cosmic layer	Cosmic structure

G.3 Splitting of K

$$\Delta K_m = 2.164 \times \Delta K_{m-1}$$

Detailed derivation see V1 Volume 18, V3 Chapter 12.

G.4 Resonance of K

Two structures share K if and only if their frequency ratio is rational:

$$r_{AB} = \frac{m}{n}$$

Resonance condition: $r_{AB} = 24^{\Delta K}$.

Appendix H Interpretation of 2/137 in the Proton Formula

H.1 Formula

$$M_p = M_e \times 24^2 \times \pi \times \left(1 + \frac{2}{137}\right)$$

H.2 Source of 2/137

The proton outer skin has two faces:

- Inner surface: facing the proton interior, the interface between quarks and corner points;
- Outer surface: facing the yang-side light sea, the interface between the proton and the light sea.

Each face contributes one fine-structure constant $\alpha = 1/137$. Therefore, the coefficient is $2/137$.

H.3 Geometric Source of the Fine-Structure Constant

$$\alpha^{-1} = 40.5 \times \frac{500}{2025} \times 15.55 \times 0.884 = 137.5$$

where $40.5 = 9^2/2$ is the total number of channels, $500/2025$ is the odd-even meeting weight, 15.55 is the wave-propagation modulation, and 0.884 is the quantum correction.

H.4 Outer-Skin Energy

$$E_{\text{outer skin}} = M_{\text{bare}} \times \frac{2}{137} = 924.7 \times 0.0146 = 13.5_{\text{MeV}}$$

This is the origin of the 13.6 MeV in the proton mass formula.

Detailed derivation see V1 Volume 4, V3 Chapter 3.

Statement on Mathematical Language

This volume uses the standard mathematical language of fiber bundles, differential geometry, and topology to provide an analogical translation of the elementary-mathematical conclusions of Volumes I–III. Concepts such as "Chern-like quantity," "holonomy-like condition," and "intersection index" are quantities defined by the author to adapt to the non-compact base space and fractal structure of this theory; they do not fully satisfy all the conditions of the standard mathematical definitions. For all formulas involving standard mathematical definitions, their form is correct; however, their use in this theory is, in part, physical hypothesis or analogical extrapolation, not rigorous mathematical derivation.

This volume is not a rigorous mathematical treatise. It is an analogical translation of a physical framework that was originally developed in elementary mathematics. The author will not update this volume further. The rigorous mathematical formalization of the framework is left to other interested researchers.

The author clearly states: all statements in this volume labeled "theorem," "proposition," or "proof" are to be understood within the author's own axiomatic system. They are not theorems in the sense of standard mathematics, and they do not claim to be rigorous mathematical results. The author's axiomatic system consists of the geometric axioms established in Volumes I–III of this series, plus the physical hypotheses explicitly stated in this volume.

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